CHEMISTRY GRADE 10: PERIODICITY

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.5 (7 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Chemistry
Strand	Strand 1.0: Chemical Families
Sub-Strand	Sub-Strand 1.5: Periodicity
Total Duration	7 lessons (approximately 14 periods × 40 minutes = 560 minutes)
Sub-Strand Content	– Trends in groups I, II, VII, VIII and period 3 – Gradation in size of atoms and ions, appearance, ionisation energy, electron affinity, melting and boiling points – Ductility, malleability and electrical conductivity – Chemical properties of group I and II elements (reaction with oxygen, chlorine, cold water, steam and dilute acids) – Physical properties of chlorine, bromine and iodine (appearance, smell, solubility in water and physical states) – Chemical properties of chlorine (reaction with water, hydrogen, metals, displacement reactions, bleaching action) – Physical properties of period three elements (atomic size, ionisation energy, electron affinity, electronegativity, melting and boiling points) – Chemical properties of period three elements with oxygen, water, chlorine and dilute acids – Uses of selected elements in groups I, II, VII and VIII
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) describe the trends in physical properties of elements of the periodic table; b) investigate the chemical properties of elements in groups of the periodic table; c) describe the trends in properties across a period; d) outline applications of elements of the periodic table; e) appreciate applications of various elements of the periodic table.
Core Competencies	– Communication and Collaboration: The learner discusses in groups the trends in physical properties of elements in groups I, II, VII and VIII, practising respectful scientific dialogue and peer explanation — modelled on how research chemists at institutions such as the University of Nairobi collaborate to characterise new compounds. – Self-efficacy: The learner confidently communicates the uses of selected elements while making presentations in plenary, building the belief that their chemical reasoning is valid and valuable. – Learning to Learn: The learner independently practises writing balanced chemical equations for reactions involving group I, II and VII elements, taking ownership of their own conceptual gaps and filling them through self-directed revision. – Digital Literacy: The learner uses digital devices appropriately to search for information on the uses of group I, II, VII and VIII elements, critically evaluating online sources and curating relevant evidence.
Core Values	– Unity: The learner appreciates the participation of each member of the group as they carry out experiments to investigate the physical and chemical properties of elements in the chemical families, recognising that accurate results depend on every team

	member's contribution.
Science & Engineering Practices	– Asking Questions and Defining Problems: Students generate and refine questions about why sodium reacts violently with water while magnesium barely reacts, anchoring curiosity in observable evidence before any direct instruction. – Planning and Carrying Out Investigations: Learners design and conduct experiments to investigate physical and chemical properties of group I, II and VII elements — including reactions with oxygen, cold water, steam and dilute acids — recording controlled variables and safety precautions. – Analysing and Interpreting Data: Students construct data tables of periodic trends (atomic radius, ionisation energy, electronegativity, melting point) across period 3 and down groups, identifying patterns and anomalies and linking them to electron configuration. – Constructing Explanations: Learners build written and visual explanations that connect observed reactivity trends directly to changes in electron configuration, nuclear charge and shielding across periods and down groups. – Obtaining, Evaluating and Communicating Information: Students use digital and print resources to research industrial and everyday uses of selected elements in Kenya (e.g., chlorine in water treatment at Nairobi Water Company, sodium in street-light lamps along Thika Road) and present findings in plenary.
Pertinent & Contemporary Issues (PCIs)	– Learner Support Programmes (healthy inter and intrapersonal relationships): The learner searches with peers for information on the uses of elements in groups I, II, VII and VIII, building respectful collaborative relationships and healthy peer interdependence. – Socio-Economic and Environmental Issues (road safety): The learner discusses the use of selected elements — such as sodium vapour and xenon — in illuminating street lights for safe roads, connecting chemistry to public safety infrastructure in Kenyan cities like Nairobi and Mombasa. – Life Skills (assertiveness): The learner confidently presents their findings on the uses of elements to the whole class, practising assertive, evidence-based communication in a safe academic environment.
Career Connections	– Pharmacist / Pharmaceutical Chemist: Understanding periodic trends in reactivity and electronegativity underpins drug design and the selection of safe reactive compounds — relevant to Kenya's growing pharmaceutical manufacturing sector in Nairobi's industrial area. – Water Treatment Technologist: Knowledge of chlorine's chemical properties (reaction with water, bleaching, disinfection) is directly applied by technicians at water treatment plants such as those operated by Nairobi Water and Sewerage Company and the Kisumu Water and Sanitation Company. – Materials Engineer / Metallurgist: Trends in ductility, malleability and electrical conductivity across period 3 inform the selection of metals for construction and electronics manufacturing — critical skills for engineers working on Kenya's infrastructure projects. – Agricultural Soil Scientist: Understanding the reactivity of group I and II elements (e.g., potassium, calcium, magnesium) with water and acids explains soil nutrient availability for Kenyan smallholder farmers in the Rift Valley and Central Kenya. – Chemical Engineer (Industrial Chemistry): Knowledge of elemental properties and periodic trends supports process design in Kenya's chemical industries, including soda ash production at Magadi and fertiliser manufacturing.
Focus for Lessons	This unit guides Grade 10 learners to discover that the periodic table is not merely a reference chart but a predictive map of chemical behaviour — where an element's position encodes its electron configuration, which in turn determines every measurable physical and chemical property. Using a Kenyan-context anchoring phenomenon (the dramatic contrast between how lithium, sodium and potassium react with water from Lake Victoria versus how inert argon and neon are), learners build an evidence-based model of periodic trends across groups I, II, VII and VIII and across period 3. The instructional approach blends hands-on experiments, collaborative group discussion, digital research and model revision, ensuring learners move from surface-level pattern recognition to deep mechanistic understanding rooted in electron configuration.

Driving Question / Key Inquiry Questions	DRIVING QUESTION: "Why do elements that sit next to each other in the periodic table — like sodium and magnesium, or chlorine and argon — behave so differently, and can we use an element's position in the table to predict exactly how it will behave before we ever put it in a test tube?" KICD KEY INQUIRY QUESTIONS: 1. Why is the study of chemical families important? 2. How does the position of an element in the periodic table influence its properties?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Exploding Metals of Lake Victoria" A water purification officer at the Kisumu Water and Sanitation Company demonstrates that pure sodium metal — the same element found chemically bonded in ordinary table salt (sodium chloride) used daily on ugali and sukuma wiki — explodes violently when dropped into a beaker of Lake Victoria water, skimming across the surface, fizzing furiously and igniting into an orange flame, leaving behind a strongly alkaline solution. Yet in the same demonstration, a strip of magnesium (from the next group) barely fizzes in cold water, and a piece of aluminium (one step further across period 3) shows no reaction at all with cold water. Meanwhile, the sealed glass ampoule of argon gas the officer uses to shield the sodium during transport is completely inert — it reacts with nothing. Students observe: (1) dramatic, visible reactivity differences between chemically adjacent elements, (2) a clear left-to-right gradient of reactivity across period 3 that then reverses for the noble gases, and (3) striking down-group acceleration — potassium explodes more violently than sodium, which explodes more violently than lithium. WHY IT IS PUZZLING: If sodium and magnesium are neighbours on the periodic table, separated by a single proton, why is their reactivity with water so dramatically different? Why does adding one more proton (going from Na to Mg) make an element almost completely unreactive with cold water, while going DOWN group I (adding a whole new electron shell) makes the element MORE reactive? And why does argon — sitting right next to the highly reactive chlorine — refuse to react with anything at all? Students cannot yet explain these puzzles using only atomic number; they need the concept of electron configuration, shielding and ionisation energy to build a complete model.
Storyline Thread	Lesson 1 — "Something Strange at the Water Plant" (Anchoring Phenomenon + Predict Phase): Students are introduced to the anchoring phenomenon via a teacher demonstration (sodium in water) or a short video clip. In small groups they record their initial observations, generate "Notice / Wonder" statements, and make written predictions about what would happen with lithium, potassium and magnesium. The class co-constructs the initial Driving Question Board (DQB) with all unanswered questions. Sub-focus: introduction to group I trends in physical properties (atomic radius, melting point, appearance). Lesson 2 — "Mapping the Group I Family" (Observe + Explain — Group I Physical & Chemical Properties): Learners carry out a structured experiment investigating the reactions of lithium, sodium and potassium with cold water, collecting gas and testing with a glowing splint, and measuring pH of the resulting solutions. Data is tabulated and trends in reactivity, atomic size and ionisation energy are explained using electron configuration diagrams. Balanced chemical equations are written and peer-checked. Students revise their initial DQB predictions. Lesson 3 — "Magnesium, Calcium and the Stubborn Metals" (Observe + Explain — Group II Physical & Chemical Properties): Students investigate group II elements — magnesium ribbon with cold water (slow), steam (burns), and dilute hydrochloric acid (vigorous). Calcium's reaction with cold water is compared. Physical property trends (melting point, atomic radius, ionisation energy) are graphed. Learners explain why group II is less reactive than group I using a shielding and nuclear charge argument. The magnesium-burning-in-air supporting phenomenon is connected.

	Lesson 4 — "The Halogen Family: From Gas to Solid" (Observe + Explain — Group VII Physical & Chemical Properties + Chlorine Chemistry): Students examine physical samples or photographs/data cards for chlorine, bromine and iodine (appearance, physical state, smell, solubility in water). The bleaching-kanga supporting phenomenon is investigated: chlorine water is prepared and its reaction with water, metals and displacement reactions (chlorine displacing bromide, bromine displacing iodide) are demonstrated or carried out. Learners explain the trend in reactivity down group VII using electron affinity and atomic radius. Displacement reactions are linked to relative oxidising power. Lesson 5 — "The Nobles and the Patterns of Eight" (Observe + Explain — Group VIII + Period 3 Physical Properties): Students examine data for group VIII elements (helium, neon, argon, krypton) — boiling points, atomic radii, ionisation energies — and explain inertness using full outer shells. The street-light supporting phenomenon is revisited. The focus then shifts to period 3: students plot graphs of atomic radius, ionisation energy, electronegativity and melting/boiling point across Na → Ar, identify the trends and anomalies, and connect each data point back to electron configuration and nuclear charge. Ductility, malleability and electrical conductivity trends are included for the metallic elements of period 3. Lesson 6 — "Period 3 Reacts: Metals, Non-Metals and the Chlorine Test" (Investigate — Period 3 Chemical Properties): Learners carry out or observe teacher demonstrations of period 3 elements reacting with oxygen (Mg ribbon, sulfur burning, phosphorus if available), water (Na vs. Mg vs. Al), dilute hydrochloric acid (Na, Mg, Al, Si), and chlorine gas. Products are identified and equations written. Students construct a reactivity-trend summary table and answer: "How does electron configuration explain why sodium reacts explosively with water but silicon and sulfur do not?" The anchoring phenomenon is explicitly revisited and the model is refined. Lesson 7 — "What Are These Elements Good For?" (Explain + Model Building + DQB Closure — Uses and Applications): Students use digital devices or print resources to research the uses of selected elements from groups I, II, VII and VIII in Kenyan contexts (sodium in street lighting, chlorine in water treatment, argon in welding, calcium in cement/construction, iodine in medical antiseptics). Groups prepare and deliver short presentations. The class returns to the DQB and systematically answers every question posted in Lesson 1. Each student produces a final cumulative annotated periodic-table model showing how electron configuration drives the trends observed across all seven lessons, completing the storyline thread from the puzzling sodium-water explosion to a predictive, explanatory framework for all elements.
Supporting Phenomena	– Supporting Phenomenon 1 — "Bleaching a Kanga": A piece of brightly coloured kanga fabric is partially dipped into chlorine water (prepared in the school lab), and the submerged section loses its colour within minutes while the dry section remains vivid. This puzzles students: chlorine is a yellow-green poisonous gas, yet dissolved in water it becomes a bleaching and disinfecting agent used in swimming pools in Nairobi and in municipal water treatment — how does a gas become a bleaching liquid, and what is the chemistry of that transformation? – Supporting Phenomenon 2 — "The Street Light Spectrum": Photographs comparing the warm orange glow of older sodium-vapour street lights along Thika Superhighway with the blue-white glow of newer xenon/argon-filled LED street lights prompt the question: why do different group I and VIII elements emit completely different colours of light, and why is argon (a noble gas) useful precisely because it does NOT react with the hot metal filament? – Supporting Phenomenon 3 — "Magnesium Burns in a Maize-Drying Fire": A farmer in Kericho accidentally drops a strip of magnesium ribbon into a charcoal fire used to dry maize. The magnesium ignites with a blinding white flame far brighter and hotter than the charcoal, producing a white powdery ash (MgO). Students puzzle over why a shiny metal becomes a white powder, and why magnesium burns so much more intensely than the iron jembe left near the same fire. – Supporting Phenomenon 4 — "Iodine Turns Ugali Purple": A drop of iodine solution is placed on a slice of cold ugali (maize meal). The spot immediately turns dark blue-black. Yet when the same iodine solution is placed on a piece of nyama choma

	(roasted meat), no colour change occurs. This supporting phenomenon, while primarily a starch test, introduces the halogens as a group — iodine's distinctive chemistry connects to the broader question of how all halogens (chlorine, bromine, iodine) share a family resemblance yet differ systematically in reactivity and physical state.

LESSON 1 (80 minutes): Something Strange at the Water Plant	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon — the dramatically different reactivities of Period 3 elements with water — and open the Driving Question Board so that students have a personal, unresolved puzzle to investigate across all 34 lessons of the sub-strand.
Knowledge	Students will know that (1) elements in the same group of the periodic table share similar but gradually changing properties; (2) reactivity with water varies dramatically across Period 3 ($Na > Mg >> Al >> Ar \approx 0$); (3) reactivity within Group I increases down the group ($Li < Na < K$); (4) Group I metals are soft, low-density, low-melting-point solids whose physical properties show a clear downward trend.
Skills	Students will be able to (1) make structured scientific observations using a Notice / Wonder frame; (2) write and justify written predictions before instruction; (3) collaboratively formulate investigable questions and post them on a Driving Question Board; (4) construct an initial annotated model of the phenomenon using particle-level diagrams.
Attitudes	Students will appreciate that the periodic table is a powerful predictive tool, not merely a reference chart — reflecting the value of Unity as each group member's observations and questions contribute to a richer class understanding of chemical families.
Key Inquiry Question	Why is the study of chemical families important? | How does the position of an element in the periodic table influence its properties?
Purpose in Storyline	This is the opening lesson of the 34-lesson Periodicity sub-strand. It anchors all future learning in a single, puzzling real-world event — sodium exploding in Lake Victoria water — and surfaces the conceptual gaps (electron configuration, ionisation energy, shielding) that subsequent lessons will fill. Every later lesson will return to this board and this phenomenon to mark growing understanding.
Safety Notes	SAFETY: If performing a live demonstration — sodium metal is HIGHLY REACTIVE with water. Use only a pea-sized piece ($\leq 0.5\ cm^3$). Wear chemical-splash goggles, a lab coat, and heat-resistant gloves. Use a large, clear polycarbonate splash-guard in front of the trough. Keep a dry-sand bucket (NOT water) nearby for spills. Store unused sodium under mineral oil; never touch with bare hands. If showing a video instead, ensure the source is from a reputable institution (e.g., Royal Society of Chemistry or a verified Kenyan university channel). Inform students NEVER to attempt this experiment outside a supervised laboratory setting.

B. LESSON OVERVIEW
Students are introduced to the anchoring phenomenon — a water purification officer at the Kisumu Water and Sanitation Company dropping sodium metal into Lake Victoria water — through a teacher demonstration or curated video clip. Working in small groups, they record structured Notice / Wonder observations, write individual written predictions about the behaviour of lithium, potassium, magnesium, and argon, and share these with the class. The lesson closes with the co-construction of the first Driving Question Board (DQB), populated entirely with students' own unresolved questions. A brief data-analysis task on the physical properties of Group I metals (atomic radius, melting point, appearance, density) establishes the first factual sub-focus and gives students their first taste of a periodic trend before the deeper 'why' is explored in later lessons.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase	Students receive a printed	BEGIN by projecting a satellite	Strategy: INDIVIDUAL WRITTEN	OBSERVABLE INDICATOR:

📹 **VIDEO:** [Periodicity of algebraic models] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Periodicity of Chemical families] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	'Mystery Substance Card' describing two items: (A) a white crystalline powder used to season ugali and sukuma wiki every day in Kenyan homes — sodium chloride; and (B) the pure silvery metal inside that compound — sodium. Without any prior instruction, each student individually writes answers to three prediction prompts on a sticky note: (1) 'What do you think will happen if pure sodium metal is dropped into a beaker of Lake Victoria water? Describe what you see, hear, and smell.' (2) 'Will magnesium metal, sitting one step to the right of sodium on the periodic table, react the same way, more violently, or less violently? Explain your reasoning.' (3) 'Argon gas is stored right next to chlorine on the periodic table — will it react with the Lake Victoria water? Why or why not?' Students do NOT share answers yet; they seal their sticky notes and place them face-down. Pairs then briefly whisper their reasoning to each other (Think-Pair, 2 min) before the teacher takes a sample of predictions publicly.	image of Lake Victoria and a photograph of the Kisumu Water and Sanitation Company intake facility. Say verbatim: 'Every day, engineers at this plant work to make the water from Lake Victoria safe to drink. Today we are going to witness something that one of those officers encounters — and I want your brains working BEFORE you see it.' Distribute Mystery Substance Cards. Read prompt 1 aloud, then say: 'Write your prediction NOW — no talking, no searching, just your honest scientific gut.' Allow 3 minutes of silent individual writing. After writing, say: 'Turn to your elbow partner — share your prediction for prompt 2 only. You have 90 seconds. Go.' After pair share, COLD-CALL 3 students (choose 1 confident, 1 quiet, 1 who rarely volunteers): 'Amina — what did you predict for magnesium, and WHY?' Record key prediction words on the board in a T-chart labelled 'Predicted MORE reactive / Predicted LESS reactive.' Do not correct any prediction. Close by saying: 'Hold these ideas tightly — we are about to find out who was right.'	PREDICTION (Eliciting Prior Knowledge). By requiring a written commitment before any evidence is shown, this strategy activates students' existing mental models of metals and reactivity, creates cognitive dissonance when the phenomenon contradicts predictions, and gives the teacher a rapid formative snapshot of class-wide misconceptions. Posting predictions publicly (face-down, to be revealed later) creates accountability and personal investment in the outcome.	Teacher circulates during silent writing and scans sticky notes for two common misconceptions — (1) 'sodium will be safe because it is in salt' (conflating compound with element properties) and (2) 'magnesium will react the same as sodium because they are both metals.' Tally the proportion of students holding each misconception. If more than 50% hold misconception (1), make a mental note to explicitly address the compound-vs-element distinction during Phase 3. Use the cold-call T-chart data to gauge the spread of prior reasoning before proceeding.
Observe Phase 📹 **VIDEO:** [Periodicity of algebraic models] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Periodicity of Chemical families] Source: Kenya	The teacher performs a live demonstration (or plays a high-quality curated video — e.g., from the University of Nairobi Chemistry Department or the Royal Society of Chemistry) showing in sequence: (A) sodium dropped into a trough of water — students observe it skimming the surface, fizzing, igniting with an orange flame, and the resulting solution turning phenolphthalein pink; (B) a magnesium ribbon placed in cold water — barely any fizzing, no	Before starting the demonstration, distribute the Observation Table worksheet and say: 'You are scientists at the Kisumu plant. Your job is to fill in every row of this table as you watch. I will pause between each element — you will have 60 seconds to write before we move on.' SAFETY BRIEFING: Before touching any sodium, say verbatim: 'Sodium metal is stored under mineral oil because it reacts violently with moisture in the air. I am using only	Strategy: STRUCTURED OBSERVATION TABLE with GROUP COMPARISON (Evidence-Based Observation). The four-column format (See / Hear / Infer) scaffolds the NGSS Science and Engineering Practice of 'Analysing and Interpreting Data' by separating raw sensory evidence from preliminary interpretation. Group comparison immediately surfaces disagreements in the Infer column, which become productive starting	OBSERVABLE INDICATOR: During the 60-second silent writing windows, teacher scans for students who leave the 'Infer' column blank (indicates difficulty distinguishing observation from inference — a key scientific practice gap to address). After group comparison, listen for whether groups articulate a gas being produced at the sodium reaction; if fewer than half the class infer hydrogen gas production, flag this for explicit

Curriculum Tools 2025 🔍 Search ARES for similar readings	flame; (C) an aluminium strip in cold water — no visible reaction; (D) an argon-filled glass ampoule — teacher taps it, shakes it, holds it near a flame — nothing happens. Students complete a structured OBSERVATION TABLE in their exercise books with four columns: Element | What I See | What I Hear | What I Infer (a guess at what is being produced). Groups of four then compare tables and circle one observation that SURPRISED them most. Each group nominates a spokesperson to share their biggest surprise with the class.	a tiny piece — smaller than a pea. Everyone must stay behind this line. Goggles on.' Perform or play each segment, pausing for 60 seconds of silent writing after each. After all four observations: 'In your groups of four, compare your Inference column. Where do you disagree? You have 2 minutes.' After group discussion, COLD-CALL 4 group spokespeople (one per group): 'Group 3 — Juma, what surprised your group the most?' After all four share, gesture to the T-chart from Phase 1 and say: 'Look at your predictions. Without flipping your sticky note yet — show me with a thumbs-up or thumbs-down: were you surprised by what happened to magnesium?' Wait 10 seconds. Count thumbs. Say: 'Most of you predicted... Let us keep that tension in our minds.'	points for the Explain phase. Nominating one spokesperson per group develops Communication and Collaboration competency (KICD core competency).	attention in Phase 3. Collect one Observation Table per group at the end of the lesson for review.
Explain Phase 📹 **VIDEO:** Periodicity of algebraic models Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Periodicity of Chemical families Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Students receive a one-page data sheet titled 'Group I Physical Properties — A Gradient Down the Family' containing a table of Li, Na, K, Rb, Cs data: atomic radius (pm), melting point (°C), density (g/cm³), appearance description, and first ionisation energy (kJ/mol) — though ionisation energy is not yet explained; it is simply labelled 'energy needed to remove one electron (details coming in Lesson 4).' Working in pairs, students answer four guided analysis questions: (1) Describe the trend in atomic radius going from Li to Cs. (2) Describe the trend in melting point. Is the metal getting stronger or weaker as you go down? (3) A chef in Mombasa accidentally drops a small piece of lithium into a pot of water. Based on the trend you see in the data,	Distribute the data sheet. Say: 'This table holds clues about WHY the sodium demonstration looked the way it did. Work with your partner — you have 8 minutes for all four questions. Use evidence from the table in every answer.' Circulate and listen specifically for the language students use around question 3 — prompt any pair that says 'lithium would be the same' with: 'Look at the atomic radius column — what changes as you go down the group? What might that mean for how tightly the metal holds its electrons?' After 8 minutes, bring the class back. For question 1, COLD-CALL 2 students: 'Fatuma, read me your answer for question 1 — atomic radius trend.' After she answers, ask a follow-up to a second student: 'Otieno, can you add	Strategy: DATA-DRIVEN TREND ANALYSIS with GUIDED QUESTIONS (Pattern Recognition → Mechanistic Reasoning). By asking students to describe trends before explaining them, this strategy mirrors how Mendeleev himself built the periodic table — pattern first, mechanism second. The deliberate withholding of the ionisation energy explanation creates a productive knowledge gap that motivates inquiry across the next several lessons. Connecting melting point to the familiar heat of a Kisumu afternoon grounds abstract data in sensory, Kenya-relevant experience.	OBSERVABLE INDICATOR: During circulation, assess whether student answers to question 3 include specific data values (e.g., 'lithium's atomic radius is 152 pm, smaller than sodium's 186 pm, so...') or remain qualitative ('it would be less dramatic'). Students who cite specific data are demonstrating the NGSS practice of 'Using Mathematics and Computational Thinking'; flag those who do not for targeted prompting. Listen for correct use of the word 'trend' — if students say 'it changes' without direction, probe: 'Does it increase or decrease? By a little or a lot?'

	predict whether this would be more or less dramatic than the sodium explosion we just watched. Justify with data. (4) Why do you think the data sheet lists 'first ionisation energy' — what might this have to do with reactivity? (Students write a hypothesis; it will not be 'corrected' yet.) The class then shares answers to questions 1–3 in a brief whole-class discussion. Question 4 answers are posted on sticky notes to seed the DQB.	anything or challenge what Fatuma said?' WAIT 12 seconds before accepting an answer. For question 2, project the melting point column and ask: 'If I told you that caesium melts at 28°C — about the temperature of a warm afternoon in Kisumu — what does that tell you about how strongly the atoms are held together? WAIT 10 seconds. Cold-call 1 student. Transition to question 4 by saying: 'That last question — about ionisation energy — is not something I will answer today. I want your best hypothesis on a sticky note, because those questions are going straight onto our Driving Question Board.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Periodicity of algebraic models] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Periodicity of Chemical families] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Each student receives three sticky notes of different colours: BLUE = 'What I noticed that I cannot explain'; YELLOW = 'What I want to know to understand the sodium explosion'; GREEN = 'What I think connects the periodic table position to what we saw.' Students write one question or statement per note (minimum one of each colour) and bring them to a large class poster titled 'The Kisumu Water Plant Mystery — Our Growing Map of Understanding.' The teacher facilitates a 5-minute whole-class gallery walk where students read the board silently, then the class groups the sticky notes into emerging themes by voting (show of hands). Themes are circled and given provisional labels: e.g., 'Why does reactivity change across a period?', 'What is happening to electrons?', 'Why do Group I metals get MORE reactive going down?', 'What makes argon so different from chlorine?'. The Driving Question itself is written at	Distribute sticky notes and say: 'You have 3 minutes — one question or idea per note. BLUE: something you observed that you cannot yet explain. YELLOW: something you need to know. GREEN: your best guess at a connection between the periodic table and what we watched. Go.' After 3 minutes, direct students to bring notes to the poster simultaneously ('Walk up together, place your note, read two others before you sit down'). Once all notes are placed, say: 'I am going to read some of these aloud. Every time you hear a question that is ALSO in your head, tap your desk.' Read 6–8 notes, pausing after each for tapping. Cluster tapped questions together. Ask the class: 'What shall we call this cluster?' WAIT 12 seconds. Accept student-generated labels. Write the Driving Question in red marker at the top: 'Why do elements that sit next to each other in the periodic table behave so differently — and can	Strategy: COLOUR-CODED DRIVING QUESTION BOARD (Public Knowledge Construction). The three-colour system separates observations (BLUE) from knowledge needs (YELLOW) from hypotheses (GREEN), scaffolding the distinction between data, questions, and claims — core to NGSS 'Constructing Explanations.' The gallery walk and tapping ritual create a shared intellectual community around the phenomenon and gives every student, including quiet learners, a voice. Copying the DQB into exercise books externalises the class's growing model and makes progress visible across the sub-strand.	OBSERVABLE INDICATOR: Collect and photograph all sticky notes before students leave. Count the proportion of YELLOW notes that are mechanistic ('How do electrons cause reactivity?') versus descriptive ('What is the name of the gas produced?'). A high proportion of mechanistic questions indicates students are already thinking at the particle level — an indicator of readiness to move quickly into electron configuration in Lesson 2. Flag any GREEN notes that contain scientifically accurate hypotheses about electron loss to celebrate publicly at the start of Lesson 2.

	the top of the board in bold. Students copy the DQB map into their exercise books as a personal reference to update each lesson.	we use an element's position to predict exactly how it will behave before we ever put it in a test tube?' Say: 'Every lesson, we will come back here. By the end of this sub-strand, every single question on this board will have an answer — written in your handwriting, with evidence from experiments you have done.'		
Model Building Phase 📹 **VIDEO:** Periodicity of algebraic models Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Periodicity of Chemical families Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Students open their exercise books to a blank page headed: 'MY CURRENT MODEL — Lesson 1: Why does sodium explode in Lake Victoria water?' Each student draws a particle-level diagram (atoms represented as labelled circles) showing: sodium metal, magnesium metal, water molecules, and the products they infer from observation (a gas, an alkaline solution). They annotate their diagram with at least THREE labels from what they observed and at least ONE question mark where they are unsure. Below the diagram they write a 2-sentence 'Current Thinking Statement': 'Right now I think sodium reacts more violently than magnesium because... I am not yet sure why...' These models are NOT assessed for correctness — they are baseline artefacts that students will physically revise (using a different colour pen) at the start of each subsequent lesson as their understanding grows.	Say: 'Scientists never wait until they know everything to build a model. Your model today will be incomplete — and that is exactly right. I am not looking for correct answers; I am looking for HONEST thinking. Open to a new page. You have 6 minutes to draw and annotate.' Circulate and give specific verbal praise for annotations that show particle-level thinking: 'Wanjiku, I love that you drew the sodium atom and asked a question mark about what is inside it — that question will be answered in Lesson 4.' Do NOT correct wrong models — instead say: 'Interesting idea — put a question mark next to that, and let us see if the evidence changes your mind.' At the 5-minute mark, ask students to write their 'Current Thinking Statement.' COLD-CALL 2 students to read their statements aloud. After the second student shares, say: 'Notice how both statements end with uncertainty. That uncertainty is not a weakness — it is exactly what drives scientific inquiry. Clip this page — we will add to it every lesson.' Collect books to review models overnight.	Strategy: ANNOTATED PARTICLE MODEL with CURRENT THINKING STATEMENT (Progressive Model Revision). This strategy implements the NGSS Science and Engineering Practice of 'Developing and Using Models.' By establishing an explicit baseline model in Lesson 1 — with built-in question marks — students create a tangible artefact of their prior understanding that can be visibly revised with a different pen colour in future lessons. The 'Current Thinking Statement' sentence frame ('I think... because... I am not yet sure why...') separates evidence-based claims from unresolved uncertainty, developing epistemic humility alongside scientific reasoning.	OBSERVABLE INDICATOR: Overnight review of exercise books — look for three indicators of readiness for Lesson 2: (1) Does the student distinguish between the sodium atom and sodium chloride compound in their diagram? (2) Does the student's diagram show any representation of what is inside the atom (even if incorrect)? (3) Does the 'Current Thinking Statement' include a causal claim ('because') or only a descriptive one? Students who include causal language in lesson 1 are ready for deeper mechanistic reasoning in Lesson 2. Provide written feedback on 5–6 books; share anonymised examples (with permission) at the start of Lesson 2.

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) PHENOMENON LANDING — Did the sodium demonstration (or video) produce genuine surprise and curiosity, or did students seem detached? If detached, consider whether the Kisumu Water Plant framing made the context feel real and relevant — could you invite a local water officer as a guest speaker in a future lesson to strengthen the connection? (2) PREDICTION QUALITY — Were student predictions entirely random, or did they reveal structured prior knowledge about metals? What is the most common misconception you need to address first — compound vs element confusion, or the assumption that all metals behave identically? Plan a 3-minute misconception address at the start of Lesson 2. (3) DQB RICHNESS — How many of the sticky note questions are mechanistic ('Why does adding one proton change everything?') versus surface-level ('What is the orange flame?')? The ratio tells you how much scaffolding will be needed for students to engage productively with electron configuration in Lessons 3–5. (4) MODEL BASELINE — From your overnight book review, what proportion of students attempted a particle-level diagram vs a macroscopic drawing? Students who drew only beakers and flames (macroscopic) will need explicit scaffolding on the difference between macroscopic observation and particle-level explanation before Lesson 3. (5) ENGAGEMENT EQUITY — Were contributions to the DQB spread across genders and ability levels, or dominated by a few students? If dominated, adjust Phase 4 in Lesson 2 by using random cold-calling (lolly sticks with names) rather than volunteering.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students watched sodium metal dropped into Lake Victoria water — it skimmed the surface, fizzed violently, ignited with an orange flame, and turned the water alkaline (pink with phenolphthalein). Magnesium in the same water barely fizzed. Aluminium showed no reaction. Argon in its sealed ampoule was completely inert.
What did I learn?	Group I metals (Li, Na, K) show clear trends in physical properties going down the group: atomic radius increases, melting point decreases, and density generally increases. Reactivity with water also increases down the group (Li < Na < K). Going across Period 3 left to right, reactivity with cold water decreases sharply (Na > Mg > Al > Ar ≈ 0). These patterns suggest that position in the periodic table is strongly connected to behaviour — but the class does not yet know WHY.
How does this explain the phenomenon?	The class cannot yet fully explain why adjacent elements behave so differently. Current thinking is that it has something to do with what is inside the atom — specifically how electrons are arranged — but the mechanism (electron configuration, shielding, ionisation energy) has not yet been introduced. All unanswered questions are recorded on the Driving Question Board and will be addressed across subsequent lessons.

LESSON 2 (40 minutes): Mapping the Group I Family	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the physical and chemical properties of Group I elements and explain the observed reactivity trends using electron configuration and ionisation energy
Knowledge	Learners will know that: (1) Group I metals (Li, Na, K) all react with cold water to produce a metal hydroxide and hydrogen gas; (2) reactivity increases down Group I as atomic radius increases, shielding increases and ionisation energy decreases; (3) each Group I element has one electron in its outermost shell, which is released during reaction; (4) balanced symbol equations can represent these reactions including state symbols
Skills	Learners will be able to: carry out a structured experiment comparing the reactions of Li, Na and K with cold water; collect and test hydrogen gas using a glowing splint; measure pH of resulting solutions; draw electron configuration diagrams for Li, Na and K; write and balance chemical equations for Group I metal reactions with water; tabulate data and identify trends
Attitudes	Learners will appreciate that careful, systematic investigation — not guesswork — is how chemists build reliable models; they will demonstrate patience and precision in recording data, and show responsibility when handling reactive metals by following safety protocols
Key Inquiry Question	Why is the study of chemical families important? How does the position of an element in the periodic table influence its properties?
Purpose in Storyline	In Lesson 1 students observed sodium exploding in Lake Victoria water and made predictions about lithium and potassium. This lesson delivers the experimental evidence that either confirms or refutes those predictions. By directly observing and comparing Li, Na and K with water, learners gather the first concrete data set that begins to answer the driving question — the reactivity difference between neighbours on the table is rooted in electron configuration and ionisation energy, not mere chance.
Safety Notes	CRITICAL SAFETY REQUIREMENTS: (1) Lithium, sodium and potassium must be stored under oil and cut with a dry spatula/scalpel by the teacher only — students must never handle these metals directly. (2) Use only rice-grain-sized pieces of each metal. (3) Conduct all reactions in a large clear plastic trough or a large beaker placed inside a tray, behind a safety screen. (4) Potassium reaction must be done as a teacher demonstration only — students observe from at least 1.5 metres away. (5) Students must wear safety goggles throughout. (6) The resulting alkaline solutions are irritants — dispose of by diluting with excess water down the drain. (7) Never return unused metal to the stock bottle — dispose of as directed by the teacher. (8) Ensure fire extinguisher is accessible. (9) Ventilate the laboratory.

B. LESSON OVERVIEW

Learners begin by revisiting their Lesson 1 predictions about lithium and potassium. They then carry out a structured practical in which the teacher demonstrates the reaction of Li, Na and K with cold water (K as teacher-only demo). Students collect hydrogen gas over water and test it with a glowing splint, measure the pH of the resulting solutions using universal indicator or a pH meter, and record all observations in a structured results table. In the Explain phase, they draw electron configuration diagrams for Li, Na and K, link increasing atomic radius and shielding to decreasing ionisation energy, and write balanced equations with state symbols. They revise their DQB predictions and update their personal science notebooks. The lesson ends with a model-building sketch connecting electron configuration to the reactivity trend observed at the Kisumu water plant.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elements of a Polynomial Graph (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students open their science notebooks to the predictions they recorded in Lesson 1 about what would happen if lithium and potassium were dropped into water. Working in pairs for 3 minutes, they re-read their prediction and decide: do they still agree with it, want to modify it, or were they completely unsure? Each pair writes a one-sentence revised prediction on a sticky note using the sentence stem: 'I predict that [element] will react [more / less / equally] vigorously than sodium because...' Sticky notes are posted on the class Driving Question Board under the column labelled 'Our Predictions — Lesson 2'.	Teacher opens by holding up a small sealed vial of sodium (stored under oil) and saying: 'Last lesson you watched the water officer in Kisumu drop sodium into Lake Victoria water and the result was dramatic. You all made predictions — today we are going to test them with real evidence.' Teacher then reads two contrasting student predictions from Lesson 1 aloud (anonymously) without commenting on which is correct: 'One of you wrote that potassium will react MORE violently than sodium because it is lower in the group. Another wrote that potassium will react LESS violently because it is heavier and harder to move. Which prediction does the evidence support? Let us find out.' [WAIT TIME: 20 seconds of silence — do not affirm or correct yet.] Teacher reminds students: 'A prediction is not a guess — it is a claim that evidence will either support or challenge. Keep your prediction in front of you as you observe today.'	Prediction revision using the 'I used to think... now I think...' frame recorded in notebooks. Posting sticky-note predictions on the DQB makes student thinking public and creates a record that can be revisited after the experiment.	Teacher circulates and reads sticky notes. Look for: (1) whether students reference atomic structure or group position as a reason (showing transfer from Lesson 1), or (2) whether they rely only on physical appearance. Flag two or three interesting predictions to reference during the Explain phase.
Observe Phase 📹 **VIDEO:** Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elements of a Polynomial Graph (PLIX)	Students are arranged in groups of four around their lab benches. The structured practical proceeds in three stages: STAGE 1 — Lithium with water (student-conducted with teacher at the front): Each group receives a large clear plastic trough half-filled with distilled water, a small piece of universal indicator paper (or 3 drops of universal indicator solution added to the water), and a test tube for gas collection. The teacher uses tongs to place a rice-grain piece of	Before Stage 1: 'Before I give each group the lithium, I want you to write down — in the Observations column — what you PREDICT you will see for lithium based on what you already know. Leave the actual observation blank for now.' [WAIT TIME: 30 seconds for writing.] During Stage 1, circulate and prompt with: 'What is happening to the surface of the lithium? Is it floating or sinking? What does that tell you about its density relative to water?' During	Structured observation table forces systematic data recording. The predict-then-observe sequence within the table makes the discrepancy between prediction and observation visible. Comparing lithium and sodium side-by-side in real time builds the empirical trend before any explanation is offered.	Teacher checks that each group has complete data in the results table before moving to the Explain phase. Listen for misconceptions: 'the bigger piece would react faster' (confounding size with reactivity) or 'potassium is not really a metal because it floats' (incomplete metal concept). Address these briefly: 'What is the definition of a metal? Does floating on water change that definition?'

Source: CK-12 🔍 Search ARES for similar readings	lithium into each group trough. Students observe and record: speed of reaction, appearance of the metal surface, sound, whether the metal sinks or floats, gas produced (test tube inverted over metal to collect gas — teacher brings glowing splint to each group for testing), and colour change of indicator. STAGE 2 — Sodium with water (student-conducted, repeating Lesson 1 observations now with data collection): Teacher provides each group with a rice-grain piece of sodium. Students repeat observations and gas test, and measure pH using the indicator. STAGE 3 — Potassium with water (TEACHER DEMONSTRATION ONLY, students observe from seats): Teacher moves to the front demonstration bench, places a safety screen, and drops a rice-grain piece of potassium into a large water trough. Students observe the characteristic lilac/purple flame, rapid movement across the surface and vigorous fizzing from their seats. Students record all observations in a structured results table with columns: Element | Appearance of metal | Reaction speed (slow/medium/fast/explosive) | Gas produced (yes/no) | Splint result | pH of solution | Other observations.	Stage 2, ask: 'Compare your sodium observation to your lithium observation right now — what is different? Write that comparison in your notebook.' During Stage 3 (K demo), after the reaction: 'Describe what you just saw using only words — no evaluation words like amazing or scary, only what your senses detected.' [WAIT TIME: 45 seconds for silent writing before any class discussion.] Safety narration during demo: 'I am using a piece no larger than a grain of rice. Potassium is stored under oil because even contact with air moisture begins to react. This is why the Kisumu officer stored sodium under oil in that sealed container.'		
Explain Phase 📹 **VIDEO:** Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos	Students remain in their groups for 12 minutes of structured explanation activity: ACTIVITY 1 — Electron configuration diagrams (5 minutes): Each student draws the electron configuration of Li (2,1), Na (2,8,1) and K (2,8,8,1) in their notebooks side by side, labelling the nucleus, each	After Activity 1 drawings are complete, teacher selects one student to hold up their diagram for the class and asks: 'Everyone — look at the outer shell of lithium, sodium and potassium in this diagram. What single thing do they all share?' [WAIT TIME: 20 seconds.] After responses: 'Exactly	The electron configuration diagram serves as a visual anchor for the abstract concept of shielding. The bar chart of ionisation energies makes the trend quantitative rather than just qualitative. Peer-checking equations develops metacognitive accuracy. Explicit connection to the Kisumu phenomenon keeps	Collect one equation set per pair (written on half a sheet of paper) for quick marking. Exit check question (oral, whole class show-of-hands): 'Raise your hand if you can state in one sentence why potassium is more reactive than lithium.' Cold-call two students — one who raised a hand and one

📖 **HTML:** [Elements of a Polynomial Graph (PLIX)] Source: CK-12 🔍 [Search ARES for similar readings]	electron shell and the outer electron. They then answer two guided questions: (a) What do lithium, sodium and potassium have in common in their electron configurations? (b) How does the distance of the outer electron from the nucleus change as you go from Li to Na to K? ACTIVITY 2 — Linking configuration to ionisation energy (3 minutes): Students read a data strip card (provided by teacher) showing the first ionisation energies: Li = 520 kJ/mol, Na = 496 kJ/mol, K = 419 kJ/mol. They plot these three values as a bar chart in their notebooks and answer: 'What is the trend in ionisation energy going down Group I? How does this explain the reactivity trend you observed today?' ACTIVITY 3 — Writing balanced equations (4 minutes): Students write and balance the three equations — (a) $2Li(s) + 2H_2O(l) \rightarrow 2LiOH(aq) + H_2(g)$; (b) $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$; (c) $2K(s) + 2H_2O(l) \rightarrow 2KOH(aq) + H_2(g)$ — and peer-check with a partner using a checklist: (i) formulae correct, (ii) balanced, (iii) state symbols included, (iv) arrow direction correct. The teacher connects back to the phenomenon: 'The NaOH produced in the Kisumu demonstration is what made that solution strongly alkaline — that is exactly what you measured today with your universal indicator.'	— one electron in the outer shell. That single electron is what each metal gives away when it reacts. Now — which atom holds that outer electron most tightly, and why?' [WAIT TIME: 30 seconds.] Guide students toward the idea of shielding: 'Potassium has more inner shells between the nucleus and the outer electron. Those inner shells act like a shield — they reduce the pull of the nucleus on the outer electron. So potassium loses its outer electron more easily than sodium, which loses it more easily than lithium. Less energy needed to remove the electron means more reactive.' During peer-checking of equations, move around and listen. If a group has not included state symbols, say: 'Your equation tells us WHAT reacts but not in what form. A solid metal and a liquid are very different. How do we show that difference?' Reinforce Kenya context: 'The LiOH, NaOH and KOH produced in these reactions are all strong alkalis used in manufacturing soap — sodium hydroxide is used in the production of bar soap across Kenya.'	the explanation grounded in the observable event from Lesson 1.	who did not — to check depth of understanding. Target misconception to probe: 'Potassium is more reactive because it has more protons' — redirect to the outer electron distance and shielding argument.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Naming ions and	Students return to the class DQB constructed in Lesson 1. Each group reviews the questions their class posted last lesson. For 3 minutes, each group selects ONE question on the DQB that today's	Teacher directs attention to the DQB: 'Our board from Lesson 1 had many unanswered questions. Look at the questions your class posted. Which ones can we now partially answer with today's	The DQB functions as a living argument map — questions from Lesson 1 get answered with evidence from Lesson 2, while new questions scaffold into Lessons 3 and 4. This shows	Teacher photographs the updated DQB at the end of each lesson to track the progression of student thinking across all seven lessons. Check whether the answers students post are evidence-based

ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elements of a Polynomial Graph (PLIX) Source: CK-12 🔍 Search ARES for similar readings	experiment has provided evidence to answer (even partially), and writes a brief evidence-based answer on a different-coloured sticky note, attaching it directly below the original question. They also add any NEW questions that today's lesson raised — for example: 'If Group I elements all have one outer electron, what happens if there are TWO outer electrons — like in magnesium?', 'Why does adding one more proton in magnesium make it so much less reactive than sodium?', 'What makes noble gases different from all these reactive metals?'	evidence?' After groups post their answers: 'I can see several groups have answered the question about why reactivity increases down the group. But I notice no one has yet answered this question over here — [points to the question about magnesium or chlorine from Lesson 1]. Keep that question in your mind. That is exactly where we are going in Lesson 3.' Teacher reads aloud any new questions students have added and validates them: 'These are excellent scientific questions. A chemist who can ask a precise question is already halfway to an answer.'	students that science is a progressive, cumulative process of inquiry, not isolated lessons.	('We saw that K reacted faster and our data shows pH was highest for K') versus opinion-based ('We think potassium is more reactive'). Prompt for evidence language if needed.
Model Building Phase 📹 **VIDEO:** Naming ions and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elements of a Polynomial Graph (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In the final 4 minutes, each student individually draws a simple 'Model Sketch' in their notebook — a diagram showing the three Group I elements arranged vertically (Li at top, K at bottom), with their electron configuration diagrams alongside, arrows indicating the direction of increasing reactivity and ionisation energy, and a one-sentence caption: 'Reactivity increases down Group I because [student completes in own words].' Students also write a one-sentence connection to the phenomenon: 'This explains the Kisumu water plant observation because...' These model sketches are NOT collected — they remain in the student notebook as a cumulative model that will be added to in every subsequent lesson until Lesson 7.	Teacher models the format quickly on the board: 'Here is what a model sketch looks like — it is not just a picture, it is a picture that EXPLAINS something. Every arrow, every label must mean something. If your caption just says the word reactive without explaining what makes it reactive, it is not a model — it is a description.' After 3 minutes of student work: 'Swap notebooks with the person next to you. Read their caption. Does it mention electron configuration, shielding or ionisation energy? If not, ask them to add one of those ideas.' [WAIT TIME: 60 seconds for peer reading and feedback.] Close the lesson by looking ahead: 'Next lesson we investigate magnesium and calcium — Group II. You already know that magnesium barely fizzed in the Kisumu demonstration while sodium exploded. Your model should start making you wonder — if magnesium also loses electrons when it reacts, why is it so much less reactive? What is different	The cumulative model notebook sketch externalises student thinking and creates a longitudinal record of model revision across all seven lessons. Peer feedback on captions builds precision in scientific language. The forward-looking prompt creates anticipatory curiosity for Lesson 3, sustaining the storyline momentum.	Teacher does a rapid 'gallery walk' of 6 to 8 notebooks while students do peer feedback, checking that model sketches include: electron configuration diagrams, directional trend arrows, a cause-effect caption referencing atomic structure, and a sentence linking to the Kisumu phenomenon. Note any students whose captions are purely descriptive — these students need targeted support at the start of Lesson 3.

		about its electron configuration that changes everything?'		

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did students make a genuine link between electron configuration and reactivity, or did they simply memorise the trend without explanation? Evidence: listen to the language in their model captions — presence of words like shielding, ionisation energy, outer electron or nuclear charge indicates conceptual understanding, not just pattern recognition. (2) Were all students safe during the practical? Review whether any group handled metal pieces directly or failed to wear goggles — address any safety breaches at the start of Lesson 3. (3) Which student predictions from Lesson 1 were confirmed by today's evidence, and which were not? How did students respond to having their predictions contradicted? Did they revise their thinking or defend their original idea? (4) Are there students who are still struggling with the concept of electron configuration from prior learning? If so, plan a brief 5-minute review at the start of Lesson 3 using the Li/Na/K diagrams as a bridge to the Group II diagrams. (5) Did the Kenya context connections (soap manufacturing, Kisumu water plant) feel authentic to students or were they perceived as added on? Adjust the framing if needed.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Lithium reacted slowly and steadily with cold water, floating on the surface and fizzing gently. Sodium reacted faster, skimming across the surface and fizzing vigorously — consistent with the Kisumu water plant demonstration. Potassium reacted most violently, producing a lilac flame and very rapid fizzing. All three produced a gas that relighted a glowing splint, confirming hydrogen production. All three produced alkaline solutions (pH 8 to 14), with potassium producing the highest pH reading.
What did I learn?	All Group I metals have one electron in their outermost shell. Going down the group, atomic radius increases and additional electron shells increase the shielding effect, reducing the pull of the nucleus on the outer electron. This means the outer electron is more easily removed — lower ionisation energy — so the metal reacts more readily. The products of all three reactions are a metal hydroxide ($LiOH$, $NaOH$, KOH) and hydrogen gas (H_2). The alkaline solution results from the metal hydroxide dissolving in water.
How does this explain the phenomenon?	The dramatic difference in reactivity between sodium and potassium — despite both being in Group I — is explained by the increasing number of electron shells and increasing shielding as you go down the group, which progressively lowers ionisation energy. This begins to answer the driving question: position in the periodic table (specifically, which period an element occupies) determines how easily it loses its outer electron, and therefore how reactive it is. This also explains why the Kisumu water officer stored sodium in argon gas — the sodium outer electron is easily lost even to trace moisture in air, so a completely inert atmosphere is required.

LESSON 3 (80 minutes): Magnesium, Calcium and the Stubborn Metals	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate and explain the physical and chemical properties of Group II elements, and to use evidence from magnesium and calcium reactions to build a shielding-and-nuclear-charge model that explains why Group II metals are less reactive than Group I metals.
Knowledge	Learners will know: (1) the trend in physical properties of Group II elements (atomic radius, melting point, first ionisation energy) down the group; (2) the chemical reactions of magnesium with cold water, steam, dilute HCl and oxygen, and of calcium with cold water; (3) why Group II elements are less reactive than Group I elements in the same period, explained in terms of nuclear charge, shielding and the energy required to remove two electrons.
Skills	Learners will be able to: (1) carry out experiments to investigate the chemical properties of Group II elements with cold water, steam and dilute acids; (2) collect and test gases produced in reactions using a lighted splint; (3) use litmus or universal indicator to test solution alkalinity; (4) plot and interpret graphs of physical property trends down Group II; (5) write and balance chemical equations for Group II reactions; (6) construct and revise a class explanatory model linking reactivity to electron configuration.
Attitudes	Learners will appreciate: (1) the unity and collaborative participation of every group member during practical work; (2) the value of evidence-based reasoning over intuition when comparing elements that appear similar; (3) the relevance of Group II chemistry to Kenyan contexts such as limestone (calcium carbonate) in Rift Valley cement production, magnesium in fertilisers, and calcium in water hardness around Lake Victoria.
Key Inquiry Question	How does the position of an element in the periodic table influence its properties?
Purpose in Storyline	In Lessons 1 and 2 students observed the violent reactivity of Group I metals (lithium, sodium, potassium) with water and began building an electron-configuration model to explain the down-group reactivity trend. Now, in Lesson 3, students shift their gaze one column to the right: Group II. By directly comparing magnesium's sluggish cold-water reaction with calcium's vigorous one, and contrasting both with sodium's explosion from the anchoring phenomenon, learners gather the critical cross-group evidence they need — two electrons to lose versus one — to deepen their shielding and ionisation-energy model. This lesson is the pivot point: it transforms the model from a single-group trend into a two-dimensional periodic table argument (across periods AND down groups), bringing the class measurably closer to fully answering the driving question.
Safety Notes	⚠️ SAFETY NOTES — read aloud to students before practical work begins: (1) Magnesium ribbon burns with an intensely bright white flame — DO NOT look directly at the flame; look away or use tinted goggles. (2) When heating magnesium in steam, ensure the delivery tube is removed from the water BEFORE the flame is extinguished to prevent suck-back. (3) Calcium metal reacts vigorously with water and generates heat — use small pieces (~0.5 cm) only; keep water volume to 50 mL in a 100 mL beaker. (4) Dilute HCl (1 mol/L) is corrosive — wear safety goggles and gloves; wipe spills immediately with a damp cloth. (5) Dispose of all alkaline solutions by diluting with excess water before pouring down the drain. (6) Magnesium ribbon scraps must be collected in a dry container — never in water. Teacher must conduct a pre-lesson risk assessment and ensure a fire extinguisher or sand bucket is accessible.

B. LESSON OVERVIEW

Lesson 3 is an Observe-and-Explain lesson in which students gather direct experimental evidence about Group II element behaviour and use it to extend their developing periodic-table model. The lesson opens with a Predict phase in which students recall the violent sodium-water explosion from the anchoring phenomenon and predict how magnesium and calcium will behave — activating prior knowledge and creating cognitive dissonance when reality does not match expectation. In the

Observe phase, student groups run three sequential investigations: (A) magnesium ribbon in cold water, (B) magnesium ribbon burned in steam over a water trough, and (C) magnesium ribbon in dilute hydrochloric acid, followed by a teacher demonstration of calcium in cold water. Students test gases with a lighted splint and test solutions with universal indicator. They record observations on structured data tables and graph three physical properties (atomic radius, melting point, first ionisation energy) down Group II using pre-printed graph axes. In the Explain phase, groups use a 'Two-Electron Tax' sensemaking frame to articulate why removing two electrons from a Group II atom costs more energy than removing one from a Group I atom in the same period — and why going down the group (more shielding) still increases reactivity. The lesson closes with targeted Driving Question Board (DQB) updates and a cumulative model revision in which students annotate their Period 3 electron-shell diagrams with ionisation-energy arrows. Throughout, every observation is explicitly tethered back to the anchoring phenomenon: the water purification officer's sodium explosion in Kisumu, recast now as the reference benchmark against which all Group II behaviour is measured.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Worked example: Finding the formula of an ionic compound Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elements of a Polynomial Graph (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students work individually for 3 minutes, then share with a partner for 2 minutes. Each student writes predictions on a sticky note or in their exercise book: (1) 'Will magnesium react with cold water? Rank the reaction: more violent than sodium / equally violent / less violent / no reaction — and explain WHY in one sentence.' (2) 'Will calcium react more or less vigorously than magnesium with cold water? Why?' (3) 'What gas, if any, will be produced?' Students then share predictions aloud via cold-call. A class tally is written on the board: how many predict 'more than sodium', 'same', 'less', 'no reaction'. This tally stays visible throughout the lesson as a reference for later evidence comparison.	Teacher opens by replaying or verbally re-narrating the anchoring phenomenon: 'Last lesson we watched sodium skid across water in Kisumu, catch fire, and leave behind an alkaline solution. Today the water purification officer places magnesium — sodium's neighbour, just one proton heavier — into the same beaker of Lake Victoria water. What do YOU predict will happen?' WAIT TIME: 10 seconds of silence before taking any responses. Cold-call 4 students (do not ask for volunteers first): 'Amina — what did you write?' 'Juma — do you agree or disagree with Amina? Why?' 'Wanjiku — what gas did you predict?' 'Otieno — more or less violent than sodium?' Teacher records the tally on the board without affirming or correcting: 'I am not going to tell you who is right yet — that is what your experiment will tell you. Keep your predictions written down; we will return to them.'	Strategy: PREDICT-OBSERVE-EXPLAIN (POE) Activation. By forcing a written, committed prediction before any evidence is revealed, the teacher creates productive cognitive dissonance. When magnesium's cold-water reaction turns out to be barely visible (a few slow bubbles after several minutes), students experience genuine surprise — a necessary condition for deep conceptual change. The visible class tally ensures every student is invested in finding out who was right.	Observable indicator: Every student has a written prediction in their book before the practical begins. Teacher circulates and scans for students who write 'same as sodium' or 'more than sodium' — these students hold the most significant misconception (that neighbouring elements behave similarly) and should be prioritised for cold-call during the Explain phase. Teacher notes names of 2–3 such students on a clipboard.

		Key teacher phrase: 'Scientists make predictions BEFORE they observe — that is what separates an experiment from just watching something happen.'		
Observe Phase 📹 **VIDEO:** Worked example: Finding the formula of an ionic compound Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elements of a Polynomial Graph (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Students work in groups of 4. Each group runs three sequential investigations using a structured practical worksheet with pre-drawn data tables and observation prompts: INVESTIGATION A — Magnesium in Cold Water (8 min): Place a 3 cm strip of pre-cleaned magnesium ribbon into a test tube of cold water. Observe for 5 minutes. At the end, test any gas collected (if using a delivery tube and inverted test tube) with a lighted splint. Test the solution with universal indicator. Record: (a) appearance of magnesium before and after, (b) rate of bubbling (describe: vigorous / slow / barely visible / none), (c) splint result, (d) indicator colour and pH estimate. INVESTIGATION B — Magnesium Burning in Steam (8 min): Teacher demonstrates OR advanced groups set up: Heat water in a boiling tube to generate steam; hold magnesium ribbon in tongs and ignite it, then pass it through the steam. Observe the bright white flame and white ash deposit. Students record colour, brightness, and products. SAFETY: Students look away from the flame. INVESTIGATION C — Magnesium in Dilute HCl (5 min): Add a 2 cm strip of magnesium ribbon to a test	Before investigations begin, teacher reads the safety notes aloud (verbatim from the SLO safety section) and conducts a 2-minute equipment check with the class. During Investigation A, circulate to groups and prompt: 'Compare the bubble rate here to what you remember from sodium in Kisumu. How would you describe the difference — use a precise word, not just slow or fast.' WAIT TIME: 10 seconds after posing each question before accepting answers. During Investigation B (demo or group), teacher narrates: 'Notice the magnesium burns in steam — it is reactive enough when given energy, but it did not react in cold water. Why might heat change things? Hold that thought for Phase 3.' During the calcium demonstration, teacher asks: 'Watch carefully — I want three specific observations from three different people. Jomo, tell me what you see happening to the surface of the calcium. Faith, what is happening to the water? Korir, what does the splint test tell us?' Cold-call by name, WAIT TIME 10 seconds each.	Strategy: STRUCTURED OBSERVATION with COMPARATIVE RANKING. Rather than allowing students to observe each reaction in isolation, the worksheet and teacher questioning force students to continuously compare Group II reactions to each other AND to the sodium benchmark from the anchoring phenomenon. The ranking task converts qualitative observations into an ordered sequence — a proto-model — that will be explained mechanistically in Phase 3. The graphing task runs concurrently, so students are building quantitative physical-property trends at the same time as they gather qualitative chemical-property evidence, priming the connection between the two in the Explain phase.	Observable indicators: (1) Each group produces a completed data table with observations for all three investigations before moving to Phase 3 — teacher initials each table during circulation. (2) Each group places sodium correctly in the vigour ranking (between calcium and Group I extremes, or above magnesium in cold water) — if a group cannot do this, teacher prompts: 'Go back to your Lesson 1 notes — what did you record about sodium in water?' (3) All three graphs show a correct trend direction (atomic radius increases down the group; first ionisation energy and melting point decrease down the group) — teacher scans graphs during the transition to Phase 3 and flags incorrect trends for targeted correction.

	tube containing 5 mL of 1 mol/L HCl. Observe rate of reaction, temperature change (feel the outside of the tube carefully), and test gas with a lighted splint. Record all observations. TEACHER DEMONSTRATION — Calcium in Cold Water (7 min): Teacher drops a 0.5 cm piece of calcium into 50 mL of cold water in a beaker while students observe. Students test the resulting solution with universal indicator. Groups compare calcium's cold-water reaction rate to magnesium's cold-water reaction from Investigation A. GRAPHING TASK (concurrent, completed during wait times): Using the data table printed on the back of the worksheet, students plot three graphs on pre-drawn axes: (1) Atomic radius (pm) vs. Group II element (Be→Ba); (2) First ionisation energy (kJ/mol) vs. element; (3) Melting point (°C) vs. element. Data values are provided in the worksheet. Students draw best-fit lines and annotate the trend direction with an arrow.	After all investigations, teacher directs: 'In your groups, rank today's reactions from most to least vigorous: calcium in cold water, magnesium in cold water, magnesium in steam, magnesium in HCl. Then place SODIUM from last lesson's phenomenon into that ranking. Where does sodium fit? Discuss for 2 minutes — I will cold-call one reporter per group.' Key teacher phrase: 'Your data from this bench is your evidence. Scientists in Nairobi, Tokyo and London doing this same experiment would get the same ranking. That is why we trust it.'		
Explain Phase 📹 **VIDEO:** [Worked example: Finding the formula of an ionic compound] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Elements of a Polynomial Graph]	Groups spend 5 minutes writing balanced chemical equations for each reaction they observed, using a scaffold provided on the worksheet: (1) $Mg(s) + 2H_2O(l) \rightarrow Mg(OH)_2(s) + H_2(g)$ [cold water, slow] (2) $Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$ [steam] (3) $Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$ [dilute acid] (4) $Ca(s) + 2H_2O(l) \rightarrow$	Teacher introduces the 'Two-Electron Tax' frame explicitly: 'I want to give you a thinking tool called the Two-Electron Tax. Every time a Group II atom reacts, it must pay a tax of TWO electrons. Group I only pays ONE. Which tax is harder to pay?' WAIT TIME: 12 seconds. Cold-call 3 students: 'Akinyi — which is harder, and why?' 'Mwangi — does nuclear charge make that tax easier or harder to pay for magnesium compared to sodium?' 'Chebet —	Strategy: TWO-ELECTRON TAX FRAME + JIGSAW PRESENTATION. The 'Two-Electron Tax' is a mechanistic analogy that gives students a memorable, causal handle on the ionisation-energy argument without reducing it to a memorised rule. By requiring students to link the tax metaphor explicitly to their graph data (ionisation energy) and their practical data (reaction rate), the frame demands multi-representational reasoning — a	Observable indicators: (1) Each group's 3-sentence explanation correctly uses the words 'ionisation energy', 'nuclear charge' or 'shielding', and 'two electrons' — teacher reads and ticks these terms during circulation. (2) Balanced equations are checked: common errors to watch for include missing state symbols, unbalanced hydrogen (H_2 not H), and writing $Ca(OH)_2$ as CaOH. Teacher corrects equations on the board in real time. (3) Jigsaw

(PLIX) Source: CK-12 🔍 Search ARES for similar readings	$Ca(OH)_2(aq) + H_2(g)$ [cold water, faster] (5) $2Mg(s) + O_2(g) \rightarrow 2MgO(s)$ [burning in air] Groups then apply the 'Two-Electron Tax' sensemaking frame (introduced by the teacher): to react, a Group II atom must lose TWO electrons; a Group I atom only needs to lose ONE. Students annotate their electron configuration diagrams for Na and Mg: (Na: 2,8,1 — one outer electron to lose; Mg: 2,8,2 — two outer electrons to lose, and the second is held by a higher effective nuclear charge). Groups write a 3-sentence explanation: 'Magnesium is less reactive than sodium with cold water because… Going from magnesium to calcium down Group II, reactivity increases because… This connects to the Kisumu phenomenon because…' A group spokesperson presents one explanation to the class (jigsaw-style, each group takes one sentence). The class co-constructs a full paragraph explanation on the board.	so what does that tell us about reactivity?' When groups write their 3-sentence explanations, teacher circulates and listens for the key conceptual link: does the student connect ionisation energy (from the graph) to reactivity (from the practical)? Prompt if missing: 'Your graph shows magnesium has a higher first ionisation energy than sodium. What does that mean for how easily it loses an electron? What does that mean for how fast it reacts?' During the jigsaw presentation, teacher scribes key phrases on the board verbatim as students say them, then asks the class: 'Is this explanation consistent with our graph data? Thumbs up if yes, sideways if unsure.' Address any sideways thumbs immediately. Key teacher phrase: 'Connect every explanation back to the officer in Kisumu — if she had dropped magnesium instead of sodium into Lake Victoria water, what would the crowd have seen differently, and WHY?' Close the phase by returning to the class prediction tally on the board: 'Who predicted magnesium would be less reactive than sodium? Your prediction was supported by evidence. Who predicted the same or more? What was your reasoning, and where did it go wrong?' WAIT TIME: 10 seconds before cold-calling 2 students who had incorrect	hallmark of NGSS Science and Engineering Practice 6 (Constructing Explanations). The jigsaw presentation ensures all students hear and critique multiple explanation attempts, consolidating understanding through social sensemaking.	presentations are evaluated for causal language: does the student say 'because' and cite a mechanism, or only describe what happened? Students who only describe (no causal language) are prompted: 'Tell me WHY, not just WHAT.'

		predictions.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Worked example: Finding the formula of an ionic compound Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elements of a Polynomial Graph (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Each student writes two contributions on sticky notes (or in their DQB section of their exercise book): GREEN NOTE — 'Something we can now answer about the driving question': e.g., 'We can now explain why magnesium reacts less violently than sodium with water — it must lose two electrons and has a higher effective nuclear charge.' OR 'We can predict that barium will react more vigorously with cold water than calcium, because shielding increases down Group II, lowering ionisation energy.' ORANGE NOTE — 'A new question this lesson has raised': e.g., 'If shielding explains the down-group trend in Group II, does the same logic apply across Period 3 — from sodium to magnesium to aluminium?' OR 'Why does magnesium react in steam but not cold water — is it just about temperature, or does the electron structure change?' Students post notes on the class DQB (a large paper sheet or section of the board organised under 'What we can explain' and 'What we still wonder'). Teacher reads 3–4 notes aloud and explicitly links them to the driving question.	Teacher frames the DQB update: 'The driving question asks whether we can predict how an element will behave from its position in the table. Three lessons in — can we? What can we predict confidently now, and what is still out of reach?' WAIT TIME: 10 seconds of individual thinking before students write. After posting, teacher reads 2 green notes and 2 orange notes aloud, then asks: 'Which of these orange questions do you think we will be able to answer by Lesson 7? Give me a thumbs up if you think we will crack it.' Use this as a preview of the sequence ahead. Key teacher phrase: 'Every question on this board is a gap in our model. By the end of this unit, we want to fill every gap with evidence — not guesses.' Teacher explicitly bridges to the next lesson: 'Your orange notes have already started asking about Period 3 trends and about Group VII. Those are exactly where we go next. Keep those questions alive.'	Strategy: DRIVING QUESTION BOARD as CUMULATIVE KNOWLEDGE TRACKER. The DQB functions as a visible, public record of the class's growing explanatory model. Distinguishing 'what we can now explain' from 'what we still wonder' trains students in the epistemic habit of knowing the boundary of their own knowledge — a core dimension of NGSS Practice 8 (Obtaining, Evaluating, and Communicating Information). The orange notes also serve a motivational function: students see their own curiosity shaping the lesson sequence ahead.	Observable indicators: (1) Green notes cite specific evidence from today's lesson (not vague generalities) — teacher redirects students who write only 'we learned about Group II' by prompting: 'What SPECIFICALLY can you now explain that you could not explain this morning?' (2) Orange notes are genuine questions (contain a question mark and identify a specific gap) rather than restatements of lesson content. (3) Teacher checks that at least one green note per group explicitly references the anchoring phenomenon (sodium in Kisumu) as the comparison benchmark.
Model Building Phase 📹 **VIDEO:**	Students retrieve their cumulative electron-shell model diagrams (started in Lesson 1, updated in Lesson 2). They now complete	Teacher frames the model revision: 'Your model is a living document — every lesson, new evidence should change it. Today	Strategy: CUMULATIVE MODEL REVISION with PREDICTIVE EXTENSION (NGSS Practice 2: Developing and Using Models). By	Observable indicators: (1) Electron-shell diagrams for Mg and Ca are structurally correct (2,8,2 and 2,8,8,2) with outer electrons

Worked example: Finding the formula of an ionic compound Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Elements of a Polynomial Graph (PLIX) Source: CK-12 🔍 Search ARES for similar readings	three tasks on the diagram: TASK 1 — Add Magnesium and Calcium: Draw electron-shell diagrams for Mg (2,8,2) and Ca (2,8,8,2) alongside the existing Na (2,8,1) and K (2,8,8,1) diagrams. Label the number of outer electrons and the number of filled inner shells. TASK 2 — Annotate with Ionisation Energy Arrows: Draw a downward arrow labelled 'IE decreases' from Be→Ba and a rightward arrow labelled 'IE increases' from Na→Mg. Annotate: 'More inner shells = more shielding = outer electrons farther from nucleus = lower IE = MORE reactive (Group II down-group)' and 'More protons, same shells = stronger nuclear pull = higher IE = LESS reactive (Na→Mg across period)'. TASK 3 — Prediction Extension: In a box at the bottom of the diagram, students write: 'Based on our model, I predict that barium (Group II, Period 6) will [react / not react] with cold water [more / less] vigorously than calcium because…' Students complete the sentence using their model. Groups share their Prediction Extension sentences with an adjacent group and evaluate: 'Is their prediction consistent with the model? What evidence would we need to test it?'	you have two new pieces of evidence: (1) Group II metals react, but less vigorously than Group I; (2) reactivity still increases DOWN Group II despite more protons. Your model must be able to explain BOTH. If it cannot, it is incomplete.' Circulate during Task 1 and check that students draw TWO electrons in the outer shell for Mg and Ca — a common error is drawing only one. Prompt: 'Count the electrons in the outer shell for magnesium — how many does the periodic table tell us it has?' During Task 2, prompt: 'Your arrow for Na→Mg goes which direction? Ionisation energy goes up or down as we move right? What does that mean for reactivity?' WAIT TIME: 12 seconds. Cold-call 2 students. After Task 3 peer sharing, ask the class: 'Did both groups reach the same prediction for barium? If you disagreed, which model is more consistent with the evidence from today?' Cold-call one pair to share their disagreement and resolution. Key teacher phrase: 'A good model does two things: it explains what you already observed AND it makes predictions you have not tested yet. Your prediction about barium is the model doing its job.'	requiring students to update a single persistent diagram across all seven lessons, the model-building phase makes knowledge growth visible and tangible. The predictive extension (barium) is critical: it forces students to use the model generatively rather than just descriptively, operationalising the driving question ('can we predict how an element will behave before we put it in a test tube?'). The peer evaluation step introduces scientific critique as a social practice.	clearly labelled — teacher checks all diagrams before the lesson ends and marks incorrect ones with a small 'R' (revise) for students to fix at home. (2) Ionisation energy arrows point in the correct directions and are labelled with a causal explanation (not just a trend statement). (3) Prediction Extension sentences for barium use causal language referencing shielding or ionisation energy — vague predictions ('barium will react more') without a 'because' are flagged. (4) Exit check: teacher asks 3 randomly selected students (cold-call by name): 'In one sentence — why is magnesium less reactive than sodium with cold water?' Acceptable answer must include reference to two electrons and/or higher ionisation energy.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions in your professional journal:\n\n1. CONCEPTUAL DEPTH — Did students reach a causal explanation (ionisation energy / nuclear charge / shielding) or did they stay at the descriptive level ('magnesium reacts more slowly')? Which groups needed the most prompting to move from description to mechanism? What question or prompt worked best to push them deeper?\n\n2. PRACTICAL MANAGEMENT — How well did Investigation A (magnesium in cold water) work in practice? If bubbles were too slow to observe meaningfully, note whether pre-cleaning the ribbon with sandpaper was sufficient, or whether a longer observation window is needed. Were groups able to collect enough gas over cold water for a reliable splint test?\n\n3. ANCHORING PHENOMENON CONNECTION — Did students spontaneously reference the Kisumu sodium explosion as a comparison benchmark, or did they need teacher prompting to make that connection? If prompting was always needed, consider making the comparison more structurally explicit in the worksheet (e.g., a column headed 'How does this compare to sodium in Kisumu?').\n\n4. DQB QUALITY — Were the orange (wonder) notes genuine new questions raised by today's evidence, or recycled questions from Lessons 1–2? If recycled, this suggests students do not yet see today's evidence as generating new puzzles — consider spending an extra 2 minutes in Phase 4 modelling what a genuinely new question looks like.\n\n5. EQUITY CHECK — Which students were cold-called today? Review your clipboard notes: were the same students called repeatedly? Identify 2–3 students who have not yet spoken aloud in plenary and plan specific cold-call moments for them in Lesson 4.\n\n6. MODEL COMPLETENESS — Are students' cumulative diagrams building coherently across lessons, or are they becoming cluttered and hard to read? If the latter, consider providing a clean A3 template for Lesson 4 that students can use to re-draw their model with all updates to date incorporated neatly.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students observed: (1) magnesium ribbon producing very slow or barely visible bubbles in cold water, with the solution turning faintly alkaline (universal indicator green-yellow, pH ~9–10 after several minutes); (2) magnesium burning vigorously in steam with a bright white flame, producing white MgO ash; (3) magnesium reacting rapidly with dilute HCl, producing vigorous bubbling and a warm test tube, with the gas giving a squeaky pop with a lighted splint; (4) calcium reacting noticeably faster than magnesium with cold water, producing steady bubbles and a distinctly alkaline solution (pH ~12). Students graphed: atomic radius increasing down Group II (Be→Ba), first ionisation energy decreasing down Group II, and melting point decreasing down Group II.
What did I learn?	Students learned: (1) Group II metals are less reactive than Group I metals in the same period because they must lose TWO electrons rather than one, and their higher nuclear charge (more protons) makes those electrons harder to remove, resulting in a higher first ionisation energy; (2) reactivity within Group II increases DOWN the group because additional electron shells increase shielding, reducing the effective nuclear charge felt by outer electrons and lowering the ionisation energy; (3) the chemical reactions of magnesium with cold water, steam, dilute HCl and oxygen follow predictable patterns that can be expressed as balanced equations; (4) calcium's stronger reaction with cold water compared to magnesium is consistent with the down-group trend predicted by their model.
How does this explain the phenomenon?	Students explained: Using the 'Two-Electron Tax' frame, magnesium is less reactive than sodium with cold water because Mg must lose two electrons (2,8,2) while Na loses only one (2,8,1); moreover, Mg has one extra proton, increasing the nuclear charge that holds the outer electrons more tightly, raising the first ionisation energy. Going from magnesium to calcium, reactivity increases because the extra electron shell in calcium (2,8,8,2) shields the outer electrons from the nucleus more effectively, lowering the ionisation energy despite the greater nuclear charge. This explains the observed trend: calcium reacted faster than magnesium with cold water, and both reacted far less dramatically than sodium — just as the water purification officer in Kisumu would predict if she switched from sodium to a Group II metal in her demonstration at Lake Victoria.

LESSON 4 (80 minutes): The Halogen Family: From Gas to Solid	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the physical and chemical properties of the halogens (Group VII elements) and explain the trend in reactivity down the group using electron affinity and atomic radius.
Knowledge	Students will know: (1) the physical properties of chlorine, bromine and iodine (appearance, physical state, smell, solubility in water); (2) the chemical properties of chlorine — reaction with water, metals, displacement reactions and bleaching action; (3) that reactivity decreases down Group VII due to increasing atomic radius and decreasing electron affinity; (4) that displacement reactions among halogens reflect their relative oxidising power.
Skills	Students will be able to: (1) observe and record physical properties of chlorine, bromine and iodine from samples/data cards; (2) carry out or observe experiments on chlorine reacting with water and metals; (3) perform and interpret halogen displacement reactions (Cl_2 displacing Br^-; Br_2 displacing I^-); (4) write balanced chemical equations for these reactions; (5) construct an evidence-based argument linking atomic radius and electron affinity to reactivity trend down Group VII.
Attitudes	Unity: The learner appreciates the participation of each member of the group as they carry out experiments to investigate the physical and chemical properties of halogens. Self-efficacy: The learner confidently communicates the uses of selected elements while making presentations in plenary. Learning to learn: The learner independently practises writing balanced chemical equations for reactions involving Group VII elements.
Key Inquiry Question	How does the position of an element in the periodic table influence its properties? Why is the study of chemical families important?
Purpose in Storyline	In Lessons 1–3 learners established that reactivity increases DOWN Group I (Li → Na → K) because additional electron shells shield the outer electron, lowering ionisation energy. This lesson flips the question: in Group VII, reactivity DECREASES down the group. By investigating chlorine, bromine and iodine — and the dramatic bleaching of a kanga fabric by chlorine water — learners discover that halogens GAIN electrons rather than lose them, so a LARGER atom with more shielding is LESS able to attract an incoming electron. This 'mirror image' trend deepens the electron configuration model and directly advances the driving question: position in the table predicts reactivity because it encodes electron configuration, shielding and ionisation energy/electron affinity.
Safety Notes	SAFETY: Chlorine gas is toxic — all chlorine preparations and demonstrations must be performed in a well-ventilated room or under a fume cupboard. Bromine liquid is corrosive and volatile — use only pre-prepared dilute aqueous bromine solution; never handle neat bromine without gloves and eye protection. Iodine stains skin and clothing. Students must wear safety goggles and lab coats throughout. Dispose of halogen solutions in the designated waste container — do NOT pour down the sink. Teacher performs all chlorine gas demonstrations; students observe from a safe distance of at least 1.5 m. Have sodium thiosulfate solution available to neutralise any spillage.

B. LESSON OVERVIEW

Lesson 4 of 7 in the evidence-gathering sequence. Learners return to the anchoring phenomenon — the water purification officer's use of chlorine to treat Lake Victoria water and the sealed argon ampoule — and now zoom in on Group VII. In the Predict phase, learners forecast whether halogen reactivity increases or decreases down the group, drawing on the Group I trend they already know. In the Observe phase, learners examine physical samples or labelled data cards for Cl_2, Br_2 and I_2 (appearance, state, colour, solubility) and then witness or perform: (a) chlorine water bleaching a strip of kanga fabric (the supporting phenomenon); (b) chlorine water reacting with iron wool; (c) chlorine water displacing bromide ions; and (d) bromine water displacing iodide ions. In the Explain phase, learners use atomic radius and

electron affinity to construct an explanation for the decreasing reactivity trend, contrast it explicitly with the Group I trend, and link displacement reactions to relative oxidising power. The DQB is updated to record the new 'mirror-image' pattern. In Model Building, learners revise their class electron-configuration model to show how shielding affects electron GAIN (Group VII) versus electron LOSS (Group I), moving closer to a full answer to the driving question.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Valence electrons and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Chemistry of Chlorine Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Learners individually examine a printed 'Trend Card' showing Group I and Group VII side-by-side (atomic number, number of electron shells, outer electrons). They respond in writing to the prompt: 'In Group I, reactivity INCREASES down the group. Predict whether reactivity in Group VII will also increase down the group, stay the same, or decrease — and give one reason based on what you already know about electrons.' After 3 minutes of silent writing, learners share in pairs (Think-Pair-Share). Three pairs are cold-called to share contrasting predictions with the class. The teacher records all three predictions on the board under the heading 'Our Predictions — we will test these today.' Learners are then shown a photograph of a Kisumu water treatment plant with chlorine cylinders, linking back to the anchoring phenomenon: 'The same element that makes Lake Victoria water safe to drink is a member of the family we are investigating today.'	Distribute Trend Cards face-down. Say: 'Do NOT turn over until I say go.' After framing the connection to the phenomenon, say: 'Turn over. You have 3 minutes — write your prediction silently. No talking yet.' SET TIMER visibly. After 3 minutes: 'Turn to your partner — 90 seconds each to share your reasoning.' After pair talk, cold-call exactly 3 pairs using the class register (pairs 4, 11, 17): 'Pair 4 — tell us your prediction AND one reason.' WAIT 10 seconds after each response before probing. Use the probe: 'You said [student's words] — is that the same logic that explained why potassium was more reactive than sodium, or is it different?' Record each prediction verbatim on the board. Do NOT evaluate ('right' or 'wrong') — say: 'We have three different ideas on the board. Evidence from today's experiments will help us decide which prediction holds.'	Think-Pair-Share with Prediction Recording. Students first commit to a written prediction (activating prior knowledge), then test their reasoning against a peer (Communication and Collaboration competency), then hear contrasting views from three pairs. Recording all predictions publicly without evaluation creates cognitive dissonance that motivates careful observation — learners know their prediction is 'on trial' and must watch evidence closely.	Teacher circulates during silent writing and scans written predictions for two indicators: (1) Does the learner reference electron shells or the number of outer electrons (evidence of prior learning transfer)? (2) Does the learner's reasoning correctly apply the Group I logic (more shells → easier to lose electron → more reactive) and then reason BY ANALOGY or in CONTRAST for Group VII? Learners who simply guess without reasoning are noted for targeted questioning during Phase 2. Collect three representative written predictions (high/mid/low reasoning quality) to revisit during the Explain phase.
Observe Phase 📹 **VIDEO:** Valence electrons and ionic compounds Source: Khan	PART A — Physical Properties Carousel (8 min): Each group of 4 receives a sealed Physical Properties Station Kit containing: (1) a photograph card of chlorine gas in a sealed flask (yellow-	PART A: 'You have 8 minutes at your stations — complete every row of the observation table before we move on.' Circulate and ask one targeted question per group: 'What trend do you notice in	Structured Observation Table with Progressive Evidence Building. Each row of the table corresponds to one piece of evidence that will be used in the Explain phase. The deliberate sequencing — physical	Teacher checks observation tables during circulation: (1) Are students recording actual observations (colour, state change, layer colour) rather than conclusions? (2) Do students correctly identify that the

Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Chemistry of Chlorine Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	green), (2) a small sealed vial of dilute aqueous bromine solution (orange-brown), (3) iodine crystals in a sealed transparent container (grey-black solid with violet vapour). Groups complete a structured observation table: Element | Colour | Physical state at room temperature | Smell (hazard card description) | Solubility in water (data provided on card). Groups record observations and note the trend in colour intensity and physical state down the group. PART B — Chemical Properties Demonstration (17 min): Teacher performs four sequential demonstrations at the front bench while learners record in their observation tables. Demo 1 — Chlorine water + kanga strip: A 5 cm strip of brightly coloured kanga fabric (a familiar Kenyan garment) is held with tongs and dipped into freshly prepared chlorine water. Learners observe bleaching within 60 seconds. Demo 2 — Chlorine water + iron wool: Iron wool is placed in a gas jar of chlorine; learners observe brown iron(III) chloride forming. Demo 3 — Displacement 1 (Cl_2 displacing Br^-): 2 cm^3 of chlorine water is added to potassium bromide solution in a test tube; 1 cm^3 of cyclohexane is added and shaken; learners observe the organic layer turning orange-brown. Demo 4 — Displacement 2 (Br_2 displacing I^-): 2 cm^3 of bromine water is added to potassium iodide solution; cyclohexane layer turns violet. After each demo, learners are given 90 seconds to write one sentence: 'What I observed and what I think it means.' Connecting prompt displayed on board throughout: 'Each observation is a	physical state as you go from chlorine to iodine?' WAIT 10 seconds. After 8 minutes, cold-call 2 groups (groups 3 and 6) to report one observation each: 'Group 3 — just the physical states, please.' PART B: Before each demo, say: 'Watch carefully — you are looking for evidence of a chemical reaction. Write down exactly what you SEE, not what you think.' After Demo 1 (kanga bleaching): 'Turn to your partner — 45 seconds. What happened to the kanga? Where have you seen something similar at home or in your community?' Cold-call 1 pair. After Demo 3: 'The cyclohexane layer has turned orange-brown. What chemical must now be present in the organic layer?' WAIT 15 seconds. Accept responses; do not confirm yet — say: 'Hold that thought — we will explain the WHY in the next phase.' After Demo 4: Write on board: 'Cl_2 can displace Br^-. Br_2 can displace I^-. Can I_2 displace Cl^-? Write your prediction NOW — one sentence.' Cold-call 2 individuals from the register.	properties FIRST, then reactions with increasing complexity — mirrors the NGSS practice of Planning and Carrying Out Investigations: learners are not just watching but building an evidence base they own. The kanga bleaching connects abstract chemistry to a culturally familiar, materially meaningful object (a kanga is a Kenyan garment with personal and communal significance), anchoring the observation emotionally and cognitively.	organic layer colour in Demo 3 indicates displaced bromine? (3) In the written 'prediction' after Demo 4, can students extrapolate the pattern (I_2 cannot displace Cl^-)? Students who are drawing no connection between the displacement pattern and relative reactivity are flagged for a targeted question in Phase 3: 'Which halogen is the strongest oxidising agent — and how does the displacement experiment prove it?'

	clue — how does it help answer why the water officer uses CHLORINE (not bromine or iodine) to treat Lake Victoria water?'			
Explain Phase 📹 **VIDEO:** [Valence electrons and ionic compounds] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Chemistry of Chlorine] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	PART A — Constructing the Explanation (10 min): Groups receive an 'Electron Affinity Scaffold Card' showing atomic radius data for F, Cl, Br, I and a definition: 'Electron affinity is the energy released when an atom GAINS one electron to form a negative ion.' Using this card and their observation tables, groups answer three guiding questions in writing: (1) 'As you go DOWN Group VII, what happens to atomic radius?' (2) 'If the nucleus is further from the outer shell, how does this affect the atom's ability to attract an incoming electron?' (3) 'How does this explain why chlorine reacted but iodine did NOT displace chloride in the final prediction?' Groups then write a group consensus explanation (3–4 sentences) on a mini-whiteboard. PART B — Contrast with Group I (5 min): Teacher displays a split comparison board: GROUP I (loses electrons — ionisation energy decreases down group → MORE reactive) vs GROUP VII (gains electrons — electron affinity decreases down group → LESS reactive). Class discusses: 'Why is the trend in Group VII the MIRROR IMAGE of the trend in Group I?' Learners add one sentence to their whiteboard explanation connecting both groups. PART C — Equations Practice (5 min): Learners individually write balanced equations for: (1) $Cl_2(aq) + 2KBr(aq) \rightarrow 2KCl(aq) + Br_2(aq)$; (2) $Br_2(aq) + 2KI(aq) \rightarrow 2KBr(aq) +$	Distribute Electron Affinity Scaffold Cards. Say: 'You have 10 minutes in your groups — use your observation table AND this card. I want every person to write the group answer in their own notebook, not just one person.' Circulate; do NOT give answers — instead ask: 'What does a BIGGER atom mean for how tightly the nucleus can pull on an extra electron? Think about shielding.' WAIT 10 seconds after asking before offering a hint. After 10 minutes, cold-call 3 groups to read their whiteboard explanations (groups 1, 5, 8). After each: 'Does anyone want to add or challenge this explanation?' WAIT 10 seconds. Then display the split comparison board: 'Look at both groups side by side. In Group I we said MORE shells means EASIER to lose an electron. Now tell me — in Group VII, what does MORE shells mean for GAINING an electron? Pair — 60 seconds.' Cold-call 2 pairs. For equations: say 'Balanced means atoms AND charges balance. You have 5 minutes — this is individual, silent work.' Circulate; stamp correct work with a 'EVIDENCE ✓' stamp. For any student who writes an unbalanced equation, ask: 'Count the chlorine atoms on both sides — are they equal?' WAIT 10 seconds.	Scaffolded Analogical Reasoning + Whiteboard Consensus. The Electron Affinity Scaffold Card provides the data students need without giving them the conclusion — learners must construct the causal chain themselves (NGSS: Constructing Explanations). The split comparison board makes the structural analogy between Group I and Group VII explicit and visual, helping learners generalise the principle: 'Shielding and atomic radius govern reactivity in BOTH groups, but in opposite directions because one group loses and the other gains electrons.' Writing the group consensus on a whiteboard makes thinking visible and creates a shared accountability artefact. The individual equation practice embeds the KICD-mandated skill of writing balanced equations (Learning to Learn competency).	Observable indicators: (1) Whiteboard explanations must include BOTH atomic radius AND electron affinity (not just 'bigger atom = less reactive' without mechanism). (2) Students must articulate the contrast with Group I — this is the key conceptual leap of the lesson. (3) Balanced equations: all three atoms must balance; state symbols required for full marks. Teacher uses a 3-level rubric in real time: Level 3 — explanation cites both atomic radius and electron affinity with causal language ('because', 'therefore'); Level 2 — cites one factor; Level 1 — restates observation without mechanism. Groups at Level 1 or 2 receive a follow-up question: 'WHY does a larger atomic radius reduce electron affinity? What is shielding doing?'

	$I_2(aq)$; (3) $Fe(s) + 3/2\ Cl_2(g) \rightarrow FeCl_3(s)$. Teacher circulates and stamps correct equations.			
Driving Question Board (DQB) Creation 📹 **VIDEO:** Valence electrons and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Chemistry of Chlorine Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Each learner writes two sticky notes: a GREEN note ('New evidence I can now add to our explanation of the Lake Victoria phenomenon') and an ORANGE note ('A new question this lesson has raised for me'). Learners post notes on the class DQB under the column 'Lesson 4 — Halogens'. The teacher reads 4–5 notes aloud (selecting one strong GREEN evidence note, one weak GREEN note, and two ORANGE question notes). The class votes on which ORANGE question is most important for the remaining lessons to answer. The teacher then explicitly bridges: 'We now know that the water officer uses CHLORINE — not bromine or iodine — because chlorine has the highest electron affinity in the liquid/aqueous halogen range, making it the strongest oxidising agent and most effective disinfectant. This is directly predicted by Group VII's position and electron configuration. But we still cannot fully explain WHY argon — one step to the right of chlorine — reacts with NOTHING. That question stays on the board until Lesson 5.' The teacher updates the 'Evidence We Have' column on the DQB anchor poster: ✓ Group I trend explained (Lessons 1–3); ✓ Group VII trend explained (Lesson 4); PENDING: Period 3 across-period trend; PENDING: Noble gas inertness.	Distribute sticky notes. Say: 'Green note — one piece of NEW evidence from today. Orange note — one question today raised that we have NOT yet answered. You have 3 minutes, silent, individual.' After posting: Read aloud 5 notes without naming authors: 'Someone wrote… [read verbatim].' After each GREEN note: 'Is this evidence strong enough to go on our class explanation? Thumbs up, thumbs sideways, or thumbs down.' After the two ORANGE notes: 'Which of these two questions do you most want answered? Raise your hand for question A… question B…' Record the winning question on the DQB in a bright colour. Close with the explicit bridge statement above — say it slowly and point to the DQB columns as you speak: 'Group I — done. Group VII — done. What's left?' Point to the PENDING items.	Driving Question Board — Evidence Tracking + Question Generation. The DQB is a living document of the class's growing model. Distinguishing GREEN (evidence) from ORANGE (new questions) teaches students that good science generates new questions, not just answers (NGSS: Asking Questions and Defining Problems). The public voting on the most important unanswered question builds student agency and frames the remaining lessons as driven by student curiosity rather than teacher prescription.	Green sticky notes are a written formative assessment: a strong GREEN note names a specific observation (e.g., 'chlorine displaced bromide but iodine could not displace chloride') AND connects it to a mechanism (e.g., 'because chlorine has higher electron affinity'). A weak note restates only the observation. Teacher counts the proportion of strong vs. weak GREEN notes to gauge class-level conceptual understanding. If more than 40% of GREEN notes lack mechanism language, the teacher flags this as a priority for the next lesson's opening retrieval activity.
Model Building Phase	Learners return to the class cumulative model poster (begun in Lesson 1) which currently shows:	Distribute Group VII panel templates. Say: 'You have 7 minutes to add your Group VII	Cumulative Model Revision with Peer Annotation (Gallery Walk). The cumulative model poster is the	Teacher photographs completed model panels at the end of the lesson. Assessment criteria: (1)

📹 **VIDEO:** Valence electrons and ionic compounds Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Chemistry of Chlorine Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	electron shell diagrams for Li, Na, K with annotations about ionisation energy and shielding explaining increasing reactivity down Group I. Each group now adds a Group VII panel to the poster using a supplied template showing blank electron shell diagrams for Cl, Br, I. Groups annotate: (1) the increasing atomic radius down the group; (2) arrows showing the direction of decreasing electron affinity; (3) a 'reactivity arrow' pointing UPWARD (most reactive at top = Cl) with the label 'Strongest oxidising agent.' Groups also add a small 'Mini Phenomenon Connection' box: 'This explains why the Kisumu water officer uses Cl_2 to treat Lake Victoria water — it is the most reactive aqueous halogen and the strongest oxidising agent available at industrial scale.' The teacher leads a 3-minute whole-class gallery walk (groups rotate one step clockwise) where learners add one peer annotation (a tick if they agree with a neighbour's model, a question mark if something is unclear or missing). Lesson closes with the teacher posing the forward-looking question: 'We have now explained the Group I and Group VII trends using electron configuration and shielding. In Lesson 5, we move ACROSS Period 3 — from sodium to argon. Predict: will properties change gradually or suddenly as we cross the period?'	panel to the class model. Every annotation must have a REASON written next to it — not just an arrow, but a label explaining what the arrow means.' Circulate; target groups that draw reactivity arrows in the WRONG direction (common error: copying Group I direction). Ask: 'In Group I, which end of the group is MORE reactive — top or bottom? Now, in Group VII — which end?' WAIT 10 seconds. During gallery walk (3 min): 'You are reading your neighbours' model as a critical friend. Add a tick if the electron affinity annotation is correct. Add a question mark if you think something is missing or wrong.' After gallery walk: pose the forward-looking question. WAIT 10 seconds. Cold-call 3 individuals from the register for predictions. Record their predictions on the board under 'Predictions for Lesson 5.' Close: 'These predictions go onto the DQB — we will test them next lesson.'	central sensemaking artefact of the 7-lesson sequence — each lesson adds one 'layer' of explanation. By physically adding a Group VII panel alongside the existing Group I panels, learners construct the visual contrast between the two mirror-image trends. Peer annotation during the gallery walk instantiates the NGSS practice of Obtaining, Evaluating and Communicating Information: learners must evaluate a peer model against their own understanding and provide specific, actionable feedback (tick = correct, question mark = unclear/missing). The forward-looking closing question seeds cognitive preparation for Lesson 5.	Atomic radius correctly shown as increasing down Group VII (larger circles for Br and I than Cl); (2) Electron affinity arrow correctly pointing upward (Cl at top = highest); (3) Reactivity arrow correctly pointing upward (Cl = most reactive); (4) 'Mini Phenomenon Connection' box completed with a complete sentence linking the model to the Kisumu water treatment context. Any panel missing criterion 3 (most common error) is flagged — the teacher opens Lesson 5 with a targeted retrieval question for those groups.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the majority of learners correctly articulate the CONTRAST between Group I and Group VII reactivity trends — using both atomic radius and electron affinity? If not, what misconception persisted, and how will you address it in the Lesson 5 opening retrieval? (2) Were the

displacement demonstrations sufficiently clear for learners to observe the colour change in the cyclohexane layer? If cyclohexane was unavailable, did learners still identify the pattern from aqueous colour changes alone? (3) Did learners make a meaningful connection between chlorine's position (Group VII, Period 3) and its industrial use in treating Lake Victoria water — or did this remain abstract? (4) Review the GREEN sticky notes: what proportion included both an observation AND a mechanism? Adjust the Lesson 5 opening based on the ratio. (5) Communication and Collaboration (KICD core competency): did all group members contribute to the whiteboard consensus explanation, or did dominant voices take over? Plan a specific role-rotation strategy for Lesson 5 group work if this was observed. (6) Safety review: was ventilation adequate during the chlorine demonstrations? Note any adjustments for future lessons involving halogen chemistry.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed: (1) physical properties of Cl_2 (yellow-green gas), Br_2 (orange-brown liquid) and I_2 (grey-black solid with violet vapour) from samples/data cards, noting the trend in colour and physical state down Group VII; (2) a strip of brightly coloured kanga fabric being bleached within 60 seconds by chlorine water, connecting to the Kisumu water treatment phenomenon; (3) iron wool forming brown iron(III) chloride in chlorine gas; (4) chlorine water turning the cyclohexane layer orange-brown when added to potassium bromide solution (Cl_2 displacing Br^-); (5) bromine water turning the cyclohexane layer violet when added to potassium iodide solution (Br_2 displacing I^-).
What did I learn?	Learners learned that: (1) halogens exist in all three physical states at room temperature — Cl_2 (gas), Br_2 (liquid), I_2 (solid) — with colour deepening and volatility decreasing down the group; (2) chlorine reacts with water to form a bleaching solution (hypochlorous acid), explaining its use as a disinfectant in Lake Victoria water treatment; (3) reactivity in Group VII DECREASES down the group — the opposite of Group I — because atomic radius increases and electron affinity decreases, making it harder for larger atoms to attract an incoming electron; (4) displacement reactions ($Cl_2 > Br_2 > I_2$) reflect relative oxidising power and can be used to establish a reactivity series for halogens; (5) balanced equations for halogen displacement reactions require careful attention to stoichiometry and state symbols.
How does this explain the phenomenon?	Learners explained that: (1) chlorine is the most reactive aqueous halogen because it has the smallest atomic radius and highest electron affinity among the three, allowing its nucleus to attract an incoming electron most strongly — this is why the Kisumu water officer uses Cl_2, not Br_2 or I_2, to treat Lake Victoria water; (2) the Group VII trend is the MIRROR IMAGE of the Group I trend — in Group I, more electron shells make it easier to LOSE an outer electron (lower ionisation energy → more reactive); in Group VII, more electron shells make it harder to GAIN an electron (lower electron affinity → less reactive); (3) displacement reactions prove relative oxidising power: a more reactive halogen (stronger oxidising agent) will displace a less reactive halide ion from solution; (4) the anchoring phenomenon's use of argon as an inert shield (not yet fully explained) is now seen as the extreme end of a trend — as groups go right across the period, reactivity with electron gain becomes impossible for a full-shell noble gas, previewing the Lesson 5 across-period investigation.

LESSON 5 (80 minutes): The Nobles and the Patterns of Eight	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To explain the inertness of Group VIII (noble gas) elements using the concept of full outer electron shells, and to identify and explain trends in physical properties (atomic radius, ionisation energy, electronegativity, melting/boiling point, ductility, malleability and electrical conductivity) across Period 3 elements (Na → Ar) by connecting each data point to electron configuration and nuclear charge.
Knowledge	Students will know that: (1) Group VIII elements have completely filled outer electron shells (helium: 2; neon, argon, krypton: 8), which confers chemical inertness; (2) across Period 3, atomic radius decreases from Na to Cl (then increases at Ar) as increasing nuclear charge pulls electrons closer while shielding remains nearly constant; (3) first ionisation energy shows a general increase across Period 3 with characteristic dips at Al and S explained by sub-shell effects; (4) electronegativity increases across Period 3 from Na to Cl, with noble gas Ar assigned no value; (5) melting and boiling points peak at Si (giant covalent structure) and drop sharply for the non-metals; (6) Na, Mg and Al are malleable, ductile and good electrical conductors, with conductivity increasing from Na to Al due to increasing delocalised electron density.
Skills	Students will be able to: (1) plot and interpret graphs of atomic radius, first ionisation energy, electronegativity, and melting/boiling point against atomic number for Period 3; (2) identify anomalies in the ionisation energy trend (Al and S dips) and explain them using electron configuration; (3) compare Group VIII boiling point and atomic radius data and use it to infer the strength of London dispersion forces down the group; (4) construct a written evidence-based explanation linking element position to physical behaviour; (5) classify Period 3 elements as metals, metalloids or non-metals based on property data.
Attitudes	Students will appreciate that: (1) the apparent 'boringness' of noble gases (their inertness) is itself a profound chemical pattern that anchors the entire periodic table; (2) systematic data collection and graph-reading are powerful tools for revealing hidden patterns in nature; (3) unity in group work — every member contributing data points and graph sections — produces a richer, more complete picture than any individual could generate alone.
Key Inquiry Question	How does the position of an element in the periodic table influence its properties?
Purpose in Storyline	In Lessons 1–4 students gathered evidence about Group I reactivity (the Lake Victoria explosion), Group II reactivity, Group VII (halogens) properties, and electron configuration/ionisation energy as an explanatory tool. Lesson 5 now completes the survey of all four KICD-mandated chemical families by adding Group VIII, and then delivers the capstone data analysis: a full quantitative sweep of Period 3 physical properties. After this lesson, students possess every piece of evidence they need to answer the driving question — they can predict reactivity AND physical behaviour from position alone. Lessons 6 and 7 will apply this predictive power to chemical reactions and uses of elements.
Safety Notes	No hazardous chemicals are used in this lesson. All activities are data-based (graphs, tables, secondary sources). If a teacher demonstration of a noble-gas discharge tube (neon/argon street-lamp) is used, ensure the high-voltage power supply is handled only by the teacher, kept away from water, and that learners remain at least 1 metre from the apparatus. Printed data tables and graph paper are the primary materials; where digital devices are used, enforce responsible use per school ICT policy.

B. LESSON OVERVIEW
Lesson 5 of 7 in the evidence-gathering sequence opens by revisiting the sealed argon ampoule from the anchoring phenomenon: the water purification officer used argon — not nitrogen, not air — to shield the violently reactive sodium during transport. Why is argon so completely safe? Students begin by predicting what argon's

electron configuration looks like compared to sodium's, then examine a data table of Group VIII elements (He, Ne, Ar, Kr) covering boiling points, atomic radii and first ionisation energies. They use this data to explain both the inertness of noble gases (full outer shell = octet/duet rule) and the trend of increasing London dispersion forces down the group. The lesson then pivots to its central activity: a structured, collaborative graph-plotting exercise in which student groups each plot one physical-property trend across Period 3 (Na → Ar), then present their graph to the class to build a composite 'Period 3 Properties Wall'. Trends and anomalies — the dip in ionisation energy at Al, the dip at S, the conductivity ladder Na < Mg < Al — are explained using nuclear charge and electron configuration. The lesson closes with a DQB update where students explicitly record what they can now predict, and a model-building revision session connecting Group VIII inertness to the full-shell model that will anchor all subsequent chemical property explanations.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Periodicity of algebraic models Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Periodicity of Chemical families Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Students individually examine a projected image of the periodic table showing only the positions of Na (Group I, Period 3) and Ar (Group VIII, Period 3) highlighted. They write a 3-sentence prediction in their notebooks answering: 'The water purification officer stored the sodium inside a sealed ampoule of argon gas before the Lake Victoria demonstration. Using what you know about electron configuration, predict WHY argon was chosen — what is different about argon's electrons compared to sodium's, and predict what you think argon's first ionisation energy will be compared to sodium's.' Students also predict the general shape of the Period 3 ionisation energy graph (rising, falling, or irregular) by sketching a rough curve without seeing any data.	Display the phenomenon image (officer holding the argon ampoule next to the sodium jar). Say verbatim: 'Before we look at any data today, I want you to think like the officer who packed that sodium. She chose argon specifically — not air, not nitrogen, not helium from a party balloon. In your notebook, write three sentences: what is special about argon's electrons, why does that make it safe, and sketch the shape you predict for ionisation energy as we go from sodium all the way to argon.' Allow 4 minutes of individual silent writing. Then cold-call 3 students (one from different sides of the room) to share predictions. After each, say: 'Hold that prediction — we will test it against real data in the next 20 minutes.' Do NOT confirm or correct yet. Record two contrasting predictions on the board under the heading 'OUR PREDICTIONS — TO BE TESTED'.	Prediction-before-evidence (anchored prediction): By requiring students to commit a written prediction to paper before seeing data, this strategy activates prior knowledge from Lessons 1–4 (electron configuration, ionisation energy concept) and creates a cognitive stake in the outcome. When the data later confirms or surprises the prediction, the learning event is more memorable and the explanation more deeply processed.	Circulate during the 4-minute writing period. Look for: (1) students who correctly write 'Ar has 8 electrons in its outer shell / full outer shell' — these students are ready for extension; (2) students who write 'Ar has no electrons' or confuse noble gas configuration with having no electrons — flag for targeted questioning during Phase 2; (3) students whose predicted IE curve is roughly rising — acceptable baseline. Ask any student who has written nothing: 'Tell me one word that describes how sodium reacted with Lake Victoria water' — use their answer ('violent/explosive') to prompt: 'So would you store it in a jar of something equally reactive, or something that reacts with nothing?'
Observe Phase 📹 **VIDEO:** Periodicity of algebraic models Source: Khan Academy (English	Activity A (8 min) — Group VIII Data Table Analysis: In pairs, students receive a printed data card (or view on shared device) containing a table of Group VIII elements: He, Ne, Ar, Kr —	Distribute data cards and worksheets. Say: 'You have 8 minutes for Activity A — work in pairs, answer all three questions, and I want every person writing, not just one partner.' Set a visible	Jigsaw Data Assembly (Distributed Graph Construction): Each group holds one piece of the Period 3 puzzle. No single group can see the full picture until all four graphs are posted together on the	During graph plotting: check that all data points are correctly placed (circulate with an answer key). Key observable indicators: (1) Group A correctly circles Al (dip at Z=13) and S (dip at Z=16) as anomalies

- US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Periodicity of Chemical families Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	showing atomic number, electron configuration, atomic radius (pm), boiling point (°C), and first ionisation energy (kJ/mol). Pairs answer three guided questions on a worksheet: (Q1) 'Describe the trend in boiling point down Group VIII — use the word increases or decreases and link it to atomic radius.' (Q2) 'All four elements have first ionisation energies above 1000 kJ/mol. Compare this to Na (496 kJ/mol) — what does this tell you about how tightly Group VIII elements hold their electrons?' (Q3) 'Write the electron configuration of Ar. Count the electrons in its outer shell. How does this explain why the officer could safely store reactive sodium in an argon atmosphere?' Activity B (12 min) — Period 3 Graph-Plotting Relay: The class is divided into 4 groups. Each group receives a pre-printed axis sheet and a data table of Period 3 elements (Na, Mg, Al, Si, P, S, Cl, Ar) and is assigned one property to graph: Group A → first ionisation energy (kJ/mol) vs atomic number; Group B → atomic radius (pm) vs atomic number; Group C → melting point (°C) vs atomic number; Group D → electronegativity (Pauling scale) vs atomic number. Each group plots their graph on A3 paper, draws a best-fit trend line or curve, circles any anomalies, and writes a one-sentence caption. Teacher also prompts Group A and B to mark the positions of Na, Mg, Al with small metal symbols (🔩) and Cl, S, P with non-metal symbols to reinforce classification.	timer. During Activity A, circulate and ask targeted questions to flagged students: 'Look at the electron configuration of argon — 2,8,8. Count the electrons in the last shell. What is the magic number we keep coming back to?' (Wait 10 seconds.) After 8 minutes, do NOT take full answers — say: 'Keep your answers — you'll need them for the DQB later. Now move into your groups of four for the graph activity.' Assign groups and distribute A3 graph paper and data tables. Say verbatim: 'Each group is responsible for ONE graph. Your graph must have: a title, labelled axes with units, all 8 data points plotted correctly, a trend line or curve, and any anomalies circled and labelled. You have 12 minutes. At the end, your graph goes on the board — it becomes part of our class Period 3 Properties Wall.' Circulate every 2–3 minutes. For Group A (ionisation energy), prompt: 'You should see two dips — one at aluminium and one at sulfur. Before you label them as anomalies, check: what is aluminium's electron configuration? Where is its outer electron sitting?' (Wait 12 seconds before cold-calling.) For Group C (melting point), prompt: 'Silicon has by far the highest melting point. We know silicon has a giant covalent structure — does that match your graph?' For Group D, remind: 'Argon has no electronegativity value — what does that tell you?'	'Properties Wall'. This structure mirrors how real scientists divide analytical labour, promotes interdependence (Unity value), and means every student must engage deeply with at least one data set. The act of posting graphs side-by-side and then stepping back to see ALL trends simultaneously is a powerful perceptual moment — students can literally see the periodic pattern as a visual landscape.	— if not, prompt with electron configuration question above; (2) Group B shows a generally decreasing atomic radius from Na to Cl with a jump at Ar — accept this if the jump is noted and labelled; (3) Group C shows Si as the clear peak — if students draw a simple rising line, ask 'What happens to bonding structure as we go from metals to non-metals?'; (4) Group D leaves Ar blank or writes 'N/A' — if students invent a value for Ar, this is a critical misconception to address immediately.
Explain Phase 📹 **VIDEO:** Periodicity of	Part A — Gallery Walk + Anomaly Explanation (12 min): All four graphs are posted side-by-side on	Introduce the gallery walk: 'You have 2–3 minutes at each graph. GREEN sticky = I can explain this	Anomaly-as-Evidence Strategy (Green/Yellow Sticky Gallery Walk): Rather than presenting	Collect one sticky note per student at the end of the gallery walk — read yellow notes to identify the

algebraic models Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Periodicity of Chemical families Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	the classroom wall as the 'Period 3 Properties Wall'. Students do a structured gallery walk (2–3 min per graph) with sticky notes: they place a GREEN sticky note on any trend they can explain using nuclear charge or electron configuration, and a YELLOW sticky note on any anomaly or pattern they cannot yet fully explain. After the gallery walk, students return to seats and the class discusses the two most-annotated anomalies: (1) the ionisation energy dip at Al — explained by the fact that Al's outer electron is in a 3p sub-shell, slightly higher in energy and more easily removed than Mg's $3s^2$; (2) the ionisation energy dip at S — explained by electron-electron repulsion in the paired 3p orbital. Part B — Conductivity and Physical Properties of Period 3 Metals (10 min): Teacher leads a guided discussion using a data table showing electrical conductivity, malleability and ductility for Na, Mg and Al. Students record in notebooks: 'Conductivity increases Na → Al because the number of delocalised electrons per atom increases (Na: 1, Mg: 2, Al: 3).' Students then connect this back to the phenomenon: 'Sodium is the softest and least conductive of the three metals — yet it is the most reactive with water. Which property — conductivity OR ionisation energy — is the better predictor of reactivity with water?' Students write a 2-sentence answer.	using nuclear charge or electron shells. YELLOW sticky = This surprises me or I am not sure why.' After the walk, stand at the ionisation energy graph and say: 'I can see a lot of yellow sticky notes at aluminium. Let me ask — ' (cold-call a student who placed a yellow note): 'What electron configuration did we write for aluminium in Lesson 3?' (Wait 12 seconds.) Guide the class: 'Magnesium is 2,8,2 — both outer electrons are in the 3s sub-shell. Aluminium is 2,8,3 — that third electron is in a 3p sub-shell, which sits slightly HIGHER in energy. Higher energy means easier to remove. So the dip is not a mistake in the data — it is a window into sub-shell structure.' Write on board: 'ANOMALY → DEEPER PATTERN.' Repeat for the S dip: 'Sulfur is 2,8,6. Draw the 3p box diagram — phosphorus has one electron in each 3p box. Sulfur has to pair one up. Paired electrons repel each other — which one is easier to remove?' (Wait 10 seconds, cold-call.) For the conductivity discussion, say: 'Mama Achieng in Kisumu uses aluminium sufurias because aluminium conducts heat well and is malleable enough to shape. Using today's data, explain to your partner in 30 seconds why aluminium conducts better than sodium.' Cold-call 2 pairs. Close Part B with: 'Now write your 2-sentence answer about ionisation energy versus conductivity as a predictor of reactivity. We will use this in the DQB.'	anomalies as 'exceptions to memorise', this strategy frames them as the most informative data points — anomalies reveal deeper structure. By having students physically flag what surprises them BEFORE the explanation, the teacher can target explanations to actual points of confusion, and students experience the explanation as resolving their own puzzlement rather than receiving a lecture.	top 3 misconceptions before the Explain phase begins. Key indicators during discussion: (1) a student who can independently articulate 'the 3p electron in Al is easier to remove than a 3s electron in Mg' is demonstrating beyond-grade-level mastery; (2) a student who says 'ionisation energy is the better predictor of reactivity' for the final 2-sentence task is showing strong causal reasoning — ask them to justify using the Lake Victoria data from Lesson 1; (3) a student who cannot distinguish conductivity from reactivity needs a one-to-one prompt: 'Sodium conducts less than aluminium — but which one explodes in water? So conductivity and reactivity are measuring different things.'
Driving Question Board (DQB)	Students return to the class Driving Question Board (DQB). In groups of 3, they review the DQB	Say: 'Our DQB has been growing since Lesson 1 when we watched sodium explode in Lake Victoria	Progressive DQB Curation (Evidence Archaeology): The DQB functions as a living, public record	Read the evidence sentences written on the backs of moved cards — these are direct windows

Creation 📹 **VIDEO:** [Periodicity of algebraic models] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Periodicity of Chemical families] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	cards posted in Lessons 1–4 and complete two tasks: (1) Move any card from the 'We wondered…' column to the 'We can now explain…' column if today's lesson provided the answer, writing a 1-sentence evidence summary on the back of the card. Likely cards to move: 'Why is argon safe to store sodium in?', 'Why does ionisation energy generally increase across a period?', 'What makes metals good conductors?' (2) Post at least one NEW question card generated by today's lesson — likely candidates: 'Why does the melting point crash after silicon?', 'Do Period 2 elements show the same trends as Period 3?', 'Can we use electronegativity to predict what compounds an element will form?' Each group briefly shares their most important moved-card and their new question with the class.	water. Today we added a lot of new evidence. Take 5 minutes in your groups — find every card we can now answer, flip it over, and write one sentence of evidence on the back. Then add one new question that today's data raised for you.' Circulate and prompt groups who are stuck: 'Look at the argon ampoule card from Lesson 1 — can we explain that now?' After 5 minutes, call on 3 groups to share their moved card and evidence sentence. After each, say: 'Does everyone agree this question is answered? Thumbs up, thumbs sideways, or thumbs down.' Address any thumbs-down with: 'What part of the explanation are you still unsure about?' Close by pointing to the driving question displayed on the wall: 'We said at the start — can we use position in the table to predict behaviour BEFORE we put an element in a test tube? After today, I want you to think: for PHYSICAL properties, could you now make that prediction? What are we still missing for CHEMICAL properties?' (This previews Lessons 6–7.)	of the class's growing understanding across the 7-lesson sequence. Moving cards from 'wondered' to 'explained' makes learning growth visible and motivating — students see themselves becoming more capable of answering the driving question over time. The requirement to write a 1-sentence evidence summary on the back of each moved card forces condensation and consolidation of reasoning.	into student understanding. Look for: (1) sentences that cite specific data ('Ar has a first ionisation energy of 1521 kJ/mol — much higher than Na's 496 kJ/mol — so its electrons cannot be removed under normal conditions') — high quality; (2) sentences that are vague ('argon is inert because it is a noble gas') — these students have named the category without explaining the mechanism; intervene with: 'WHY is it a noble gas? What do all noble gases have in common in terms of electrons?' (3) new question cards that reference chemical properties (reactions with water, acids) — these students are connecting to the driving question and are ready for Lessons 6–7.
Model Building Phase 📹 **VIDEO:** [Periodicity of algebraic models] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Periodicity of Chemical families] Source: Kenya	Students open their cumulative atomic model (begun in Lesson 1 as a hand-drawn diagram of the Na atom reacting with water, refined in Lessons 2–4 to include electron shell diagrams, ionisation energy arrows and shielding annotations). Today they make two additions: (1) They draw a 'stability scale' bar alongside their model — a horizontal bar from 'most reactive' (K/Cs at one end) to 'completely inert' (noble gases at the other), and place Na, Mg, Al, Si, Cl and Ar on it using today's	Say: 'Every lesson we have added a new layer to our model of the periodic table. Today's layer is the PHYSICAL PROPERTY LANDSCAPE of Period 3, plus the full-shell stability of Group VIII. Open your model notebook and make two additions — I am going to draw mine on the board while you draw yours.' Sketch the stability scale live on the board, placing K at the reactive end and Ar at the inert end, and saying: 'I am placing these using ionisation energy as my ruler. High IE = hard	Cumulative Model Revision (Layered Diagram Annotation): The cumulative model is the through-line of the entire 7-lesson sequence. Each revision session forces students to integrate new evidence with prior evidence — they cannot simply draw a new diagram; they must amend an existing one, which requires them to reconcile today's data with what they already believed. The 'stability scale' addition is particularly powerful because it creates a single visual object that unifies	Collect model notebooks (or photograph models) at the end of the lesson. Success criteria for exit assessment: (1) stability scale is correctly ordered (reactive metals at one end, noble gases at the other) with at least 4 elements placed using an IE justification — MEETS STANDARD; (2) Period 3 thumbnail shows at least 3 of 4 property curves with a recognisable shape (IE rising with dips, atomic radius decreasing, melting point peaking at Si) — MEETS STANDARD; (3) the

Curriculum Tools 2025 🔍 Search ARES for similar readings	ionisation energy data as justification. (2) They add a 'Period 3 Property Landscape' thumbnail — a small hand-sketched composite of all four graphs from the gallery walk (atomic radius, IE, melting point, electronegativity) showing the overall left-to-right pattern. Each thumbnail must include an annotation arrow for at least one anomaly (the Al or S dip) with a 3-word explanation ('3p electron — easier to remove'). Students then compare their updated model with a partner and check: 'Does your model explain why the officer used argon and NOT chlorine (the element next door) to store the sodium?' Pairs discuss for 2 minutes and write one sentence.	to remove electrons = low reactivity. Low IE = easy to remove electrons = high reactivity.' Sketch the Period 3 thumbnail composite as four mini-curves stacked vertically. After 10 minutes of individual model work, say: 'Now check your model with your partner. Your model must answer this test question: WHY did the officer choose argon and not chlorine to store the sodium? Chlorine is right next to argon on the table — what does your model say?' (Wait 12 seconds.) Cold-call 2 pairs. Listen for: 'Chlorine has 7 outer electrons — it is highly reactive and would react violently with the sodium. Argon has 8 outer electrons — full shell — so it reacts with nothing.' If this reasoning is not present, say: 'Look at your stability scale — where did you place Cl? Where did you place Ar? Now what do you think?' Close the lesson: 'Next lesson we use this model to predict the CHEMICAL reactions of Period 3 elements — the last piece of evidence we need to fully answer our driving question.'	Group I reactivity (from Lesson 1), ionisation energy (Lesson 4), and Group VIII inertness (today) into one coherent explanatory framework.	argon-vs-chlorine test question is answered with reference to outer electron count — MEETS STANDARD. Students who meet all three are ready for Lesson 6. Students who fail criterion 3 (cannot distinguish Ar from Cl in terms of reactivity) need a brief 1-on-1 review of Group VIII electron configurations before Lesson 6.

D. TEACHER REFLECTION

After the lesson, reflect on the following questions and record notes for planning Lesson 6:

1. GRAPH QUALITY: Did all four groups produce graphs with correctly plotted data points, labelled axes and identified anomalies? If Group A (ionisation energy) failed to identify the Al or S dips, you will need to revisit sub-shell notation at the start of Lesson 6 before moving to chemical properties.

2. ANOMALY REASONING: During the Explain phase, were students able to articulate WHY the Al dip occurs (3p electron vs 3s electron), or did they only describe it ('it goes down at Al')? Causal reasoning about sub-shells will be essential in Lesson 6 when explaining why Al does not react with cold water but does react with steam. Note the names of 3–5 students who gave causal explanations — use them as peer-explainers in Lesson 6.

3. DQB TRANSITION: Were students able to articulate a clear boundary between 'physical properties we can now predict' and 'chemical properties we cannot yet predict

from position alone'? If students believe they can already predict all chemical properties, Lesson 6's opening prediction task (Period 3 elements reacting with water) will create productive surprise. If students feel they cannot predict anything chemical yet, affirm that ionisation energy IS the bridge — the teacher should revisit this connection explicitly at the start of Lesson 6. 4. ARGON–CHLORINE DISTINCTION: The model-building phase's test question ('why not chlorine?') is a strong litmus test for whether students have genuinely integrated Group VII and Group VIII learning, or hold them as separate, unconnected facts. If fewer than 70% of student pairs gave a correct answer citing outer electron count, plan a 5-minute warm-up for Lesson 6 that directly compares the electron configurations of Cl (2,8,7) and Ar (2,8,8) side by side before any new content begins. 5. KICD COMPETENCIES OBSERVED: Note specific instances of Communication and Collaboration (group graph work), Self-efficacy (students presenting graphs to the class), and Digital Literacy (if devices were used for data lookup). Record one example of the Unity value in action — a moment where a group's combined graphs revealed a pattern no single student's graph could show alone. Use this example in Lesson 6 as an opening affirmation. 6. PACING: This lesson covers a high volume of content (Group VIII trends AND all Period 3 physical properties). If the graph-plotting activity ran over time, consider whether Lesson 6 can open with a 5-minute 'graph check' gallery walk rather than re-teaching, to recover momentum without losing the data-analysis experience.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Students examined a data table of Group VIII elements (He, Ne, Ar, Kr) showing atomic radius, boiling point and first ionisation energy, and plotted four separate physical-property graphs across Period 3 (Na → Ar) — first ionisation energy, atomic radius, melting point and electronegativity — during a structured group graph-plotting relay, then displayed all four graphs as a 'Period 3 Properties Wall' for a class gallery walk.
What did I learn?	Group VIII elements are chemically inert because their outer electron shells are completely full (He: 2 electrons; Ne, Ar, Kr: 8 electrons), making them extremely difficult to ionise; across Period 3, atomic radius generally decreases as nuclear charge increases while shielding remains nearly constant; first ionisation energy generally increases across Period 3 with characteristic dips at Al (3p sub-shell electron) and S (paired 3p electron repulsion); melting point peaks at Si due to its giant covalent structure; electronegativity increases Na → Cl with no value for Ar; Na, Mg and Al are malleable, ductile and electrically conductive, with conductivity increasing as the number of delocalised electrons per atom increases.
How does this explain the phenomenon?	The water purification officer stored the reactive sodium in an argon atmosphere because argon's completely full outer shell (2,8,8) gives it a first ionisation energy of 1521 kJ/mol — more than three times that of sodium (496 kJ/mol) — meaning its electrons cannot be removed or shared under any ordinary chemical conditions, making argon perfectly inert and safe as a shielding gas; the dramatic reactivity difference between sodium and its Period 3 neighbours (Mg, Al) that students observed in the anchoring phenomenon can now be quantitatively explained by the steep rise in first ionisation energy from left to right across the period — each additional proton pulling the outer electrons more tightly — while the anomalies in that trend reveal the deeper sub-shell structure of the atom.

LESSON 6 (40 minutes): Period 3 Reacts: Metals, Non-Metals and the Chlorine Test	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To investigate the chemical reactions of period 3 elements with oxygen, water and dilute hydrochloric acid, construct a left-to-right reactivity trend, and use electron configuration to explain why sodium explodes in water while silicon and sulfur do not — directly answering the anchoring phenomenon.
Knowledge	Learners will know: (1) period 3 metals (Na, Mg, Al) react with oxygen forming basic oxides; non-metals (S, P) form acidic oxides; Si forms a weakly acidic oxide; (2) reactivity with water decreases sharply from Na (violent) to Mg (slow) to Al (negligible with cold water) across period 3; (3) Mg, Al and Si react with dilute HCl at decreasing rates; Na is too reactive to use safely with dilute HCl; (4) this trend is explained by increasing nuclear charge with nearly constant shielding causing higher ionisation energy, making it harder to remove outer electrons and form ions as the period is crossed.
Skills	Learners will be able to: observe and safely record reactions of period 3 elements with water and dilute HCl; write balanced chemical equations including state symbols for each reaction; plot and interpret a reactivity-trend summary table; construct and annotate an electron configuration diagram to justify each element's observed reactivity.
Attitudes	Learners will demonstrate: scientific curiosity and careful observation of reactivity differences; respect for safety protocols especially when handling sodium and dilute acids; appreciation of how a single unifying idea (electron configuration) explains a wide range of visible chemical behaviour.
Key Inquiry Question	How does the position of an element in the periodic table influence its properties?
Purpose in Storyline	Lesson 6 is the investigative climax of the evidence-gathering sequence. Having built the conceptual tools — electron configuration, ionisation energy, shielding, group trends — across lessons 2 to 5, learners now apply them to period 3 chemical behaviour. The anchoring phenomenon (sodium exploding in Lake Victoria water while magnesium barely fizzes and aluminium shows nothing) is explicitly revisited and fully explained using the complete model. This sets up Lesson 7, where the model is applied to real Kenyan uses.
Safety Notes	Sodium must be handled only by the teacher using forceps, cut under mineral oil, stored away from water sources; pieces must be small (2 mm). Dilute HCl (1 mol per litre maximum) used in a well-ventilated room or under a fume cupboard. Burning magnesium ribbon must NEVER be viewed directly — students use dark glass or look away; magnesium ash must not be inhaled. Sulfur burning produces sulfur dioxide — carry out in fume cupboard or outside; small quantities only. Wash hands thoroughly after all activities. A bucket of dry sand must be available as fire suppressant — never use water on burning sodium or magnesium.

B. LESSON OVERVIEW

Learners begin by predicting which period 3 elements will react with water and dilute HCl using ionisation energy data from Lesson 5. The teacher then demonstrates sodium, magnesium and aluminium with water and dilute HCl while students observe and record. Sulfur and magnesium burning in oxygen are demonstrated. Learners write balanced equations, construct a reactivity summary table, and use electron configuration diagrams to explain the trend. The lesson closes with an explicit return to the anchoring phenomenon: students articulate a full causal explanation for why sodium explodes in Lake Victoria water while its period 3 neighbours do not, and update the Driving Question Board.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Video: Least-Squares Regression Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Compound Interest per Period (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Students receive a Predict-Observe-Explain (POE) worksheet listing six reaction scenarios: (1) Na with cold water, (2) Mg with cold water, (3) Al with cold water, (4) Mg with dilute HCl, (5) Al with dilute HCl, (6) S burning in oxygen. For each scenario, students individually write a prediction (vigorous / slow / no reaction) and a one-sentence justification referencing ionisation energy or electron configuration from Lesson 5 data. After 3 minutes of individual thinking, pairs compare predictions and flag any disagreements. The teacher selects three pairs with different predictions to share briefly.	Display the ionisation energy bar chart from Lesson 5 on the board. Say: 'Look at the ionisation energy values we plotted last lesson — Na is 496 kJ per mole, Mg is 738, Al is 577, Si is 786. Before we touch any chemicals, use only this data to predict how each element will react with water and with dilute HCl. Write your prediction AND your reasoning — I want to see the link to electron configuration.' Allow 3 minutes of silent individual writing. [WAIT TIME: 3 minutes — resist the urge to prompt; students must commit to a prediction in writing before observing.] Then ask pairs to compare. Cold-call two pairs: 'Amara and Brian, you disagreed about aluminium — what did each of you predict and why?' Use a T-chart on the board to record the class split predictions for aluminium and magnesium only, as these are the most contested. Say: 'Let us hold these predictions. In the next phase we will find out who was right — and more importantly, why.'	T-chart of class predictions on the whiteboard makes prior knowledge visible and creates cognitive tension before observation. Requiring a written justification referencing ionisation energy ensures predictions are mechanistic, not just guesses, connecting directly to Lesson 5 learning.	Scan student worksheets during individual writing time. Look for: correct use of ionisation energy trend as a justification; any student still predicting that Al will react vigorously with cold water (common misconception — the passivating oxide layer). Note these students for targeted questioning during the Explain phase.
Observe Phase 📹 **VIDEO:** Video: Least-Squares Regression Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** Compound	Teacher carries out a structured demonstration sequence. Students complete the Observe column of their POE worksheet for each reaction, noting: what they see (colour change, gas, flame, vigour), what they hear (fizzing level), and what they can infer about products. For the water reactions: Na (teacher demo, behind safety screen), Mg ribbon in cold water (students can observe closely), Al foil in cold water (negative result — students	Begin with the anchoring phenomenon re-enactment: 'We are going back to Kisumu Water Plant. Watch this sodium — same sodium the water officer used.' Drop a 2 mm piece of sodium into a beaker of water behind the safety screen. After students react, say: 'Write exactly what you see in your Observe column — be precise. Is it more or less vigorous than Lesson 2?' [WAIT TIME: 30 seconds of silent writing.] Move through the sequence: Mg in cold	Real-time co-construction of the 4-column class table (Element / Reacts with water? / Reacts with HCl? / Reacts with oxygen?) on the board externalises observations incrementally, allowing students to check their own records against a shared reference and spot discrepancies immediately.	Use circulation during observations to check students are recording specific observations (not just 'it reacted') and are distinguishing vigour levels (violent / moderate / slow / nil). Ask individuals: 'How does the rate for Al with HCl compare to Mg with HCl? Can you quantify it?' Listen for unprompted use of the word ionisation energy — note which students are already connecting observation to mechanism.

[Interest per Period (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	record this carefully). For dilute HCl reactions: Mg ribbon (students test gas with glowing splint), Al foil (students observe rate, compare to Mg). For burning in oxygen: Mg ribbon burns brilliantly (students use dark glass), sulfur burns with blue flame producing choking gas (fume cupboard, teacher only). A 4-column observation table on the board is filled in real time as a class.	water — 'Compare the vigour to sodium. Write numbers if you can — count the bubbles per second.' Al in cold water — 'Record what you see. Nothing is also a valid scientific observation.' For HCl reactions, prompt: 'Collect the gas from the Mg reaction. What does the glowing splint tell you?' Before burning Mg, warn: 'Eyes protected — look away or use dark glass. I will count to three.' After each demonstration, pause for [WAIT TIME: 20 seconds] of individual recording. For sulfur, say from fume cupboard: 'Notice the colour of the flame and the sharp smell — that gas is sulfur dioxide. Note the product and the hazard — both are data.'		
Explain Phase 📹 **VIDEO:** [Video: Least-Squares Regression] Source: PhET Interactive Simulations (English) 🔍 [Search ARES for similar videos] 📖 **HTML:** [Compound Interest per Period (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	In groups of three, students complete the Explain column of their POE worksheet for each reaction, then tackle three constructed tasks: (A) Write balanced chemical equations with state symbols for: Na(s) + H2O(l), Mg(s) + HCl(aq), Al(s) + O2(g), S(s) + O2(g). (B) Draw electron configuration diagrams for Na, Mg, Al and Si side by side, annotating the number of outer electrons and the first ionisation energy for each, then write one sentence explaining the reactivity trend using the words nuclear charge, shielding and outer electrons. (C) Complete a reactivity summary table ranking Na, Mg, Al, Si, S, Cl, Ar from most to least reactive in period 3 chemical reactions and justify the ranking.	Say: 'Your predictions are on the board. Now explain what you observed using the electron configuration model. Do not just describe — I want a causal story: why does sodium lose its electron so easily?' [WAIT TIME: 4 minutes of group work before any teacher input.] Circulate and listen. After 4 minutes, cold-call: 'Fatuma, your group said sodium has a low ionisation energy — tell me what that actually means in terms of what is happening at the atomic level when sodium touches water.' After the response, press: 'So why does magnesium, with two outer electrons, react more slowly — surely two electrons are easier to give away than one?' [WAIT TIME: 15 seconds.] Anticipate the misconception: students may say two electrons means more reactive. Guide: 'Compare the nuclear charges of Na and Mg — Na has 11 protons, Mg has 12. With similar shielding, which ion	The side-by-side electron configuration diagram task (Task B) is the core sensemaking tool — it makes the nuclear charge versus shielding argument visual and forces students to connect the diagram directly to the number they plotted in Lesson 5. The reactivity ranking table (Task C) consolidates the whole period 3 chemical story into one artefact that will be used in Lesson 7.	Collect three to four worksheets at random and check Task B sentences for correct use of nuclear charge and shielding. Common errors to watch for: (1) student says Al is less reactive than Mg because it has more electrons — correct to nuclear charge argument; (2) student writes unbalanced equations — use peer checking to surface this. Ask exit-check oral question to the class: 'Give me one word that explains why silicon does not react with cold water.' Expect: ionisation energy or nuclear charge.

		holds its electrons more tightly?' For equation writing, display a partially balanced equation and ask: 'What is wrong with this equation?' Use peer checking — groups swap worksheets and check equations using the atom-counting rule.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Video: Least-Squares Regression] Source: PhET Interactive Simulations (English) 🔍 Search ARES for similar videos 📖 **HTML:** [Compound Interest per Period (Flexbook)] Source: CK-12 🔍 Search ARES for similar readings	The teacher projects the original Driving Question Board constructed in Lesson 1. Students individually re-read the questions they posted and, on a sticky note, write: (1) one DQB question they can now fully answer using today's evidence, and (2) one question that today's lesson raised or deepened. Students post both notes on the DQB under the headers CAN NOW ANSWER and NEW EVIDENCE NEEDED. The class reads the new additions silently for 90 seconds.	Say: 'We started this unit with a puzzling question: why does sodium explode in Lake Victoria water while magnesium barely fizzes? Look at the DQB from Lesson 1. Find the question that today most directly answers. Write the question number on your sticky note and a one-sentence answer on the back.' After posting, read two or three of the CAN NOW ANSWER notes aloud. Then say: 'Has anyone posted something under NEW EVIDENCE NEEDED? Let us hear one.' If no new questions emerge, prompt: 'We have seen that Al does not react with cold water — but the officer uses Al containers near the lake. Is Al truly unreactive or is something else protecting it? Does that belong on our board?' [WAIT TIME: 20 seconds for students to think.] Add the oxide passivation question to the DQB if raised. Preview: 'Next lesson we answer every remaining question and find out where these elements are actually used in Kenya.'	Revisiting the Lesson 1 DQB makes the learning journey visible — students see that the puzzles they could not explain in Lesson 1 now have answers grounded in evidence collected across six lessons. The dual sticky-note structure (answered versus new) models scientific thinking: answering one question always opens another.	Read the CAN NOW ANSWER sticky notes after class to assess the quality of student explanations. Look for: correct attribution of sodium reactivity to low ionisation energy and single outer electron; recognition that the trend is caused by increasing nuclear charge across the period. Identify any notes that still cite atomic mass rather than nuclear charge or electron configuration — these students need targeted feedback before Lesson 7.
Model Building Phase 📹 **VIDEO:** [Video: Least-Squares Regression] Source: PhET Interactive Simulations	Each student opens their cumulative annotated periodic table model (started in Lesson 2 and updated each lesson). For the period 3 row, they add colour-coded reactivity annotations: (1) highlight Na in red, Mg in orange, Al in yellow, Si in green, S in blue, Cl in violet, Ar in grey,	Say: 'Your annotated periodic table is your model of the whole unit. Every mark you add today is another piece of evidence that links back to the water plant in Kisumu. Add the reactivity gradient for period 3 now — make it visible on the table itself, not just in your notes.' Circulate and prompt with:	The cumulative annotated periodic table is the unit-long model-building artefact. Adding period 3 chemical reactivity data today layers chemical behaviour onto the physical property trends added in Lessons 4 and 5, producing a multi-layered predictive model. Colour-coding reactivity creates an	Check that students have drawn the trend arrow with a correct label — incorrect labels (for example: reactivity increases across the period) are a key misconception to catch before Lesson 7. Collect three to four annotated tables at the end of class to review quality of annotations and correct any

(English) 🔍 Search ARES for similar videos 📖 **HTML:** Compound Interest per Period (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	corresponding to their reactivity ranking; (2) annotate each element with its reaction outcome with water (one word: violent / slow / none) and with oxygen (burns / glows / none); (3) draw a trend arrow across the period labelled: ionisation energy increases, reactivity with water and metals decreases, with the exact data points from the Lesson 5 graph. Students who finish early extend: add the equation for the reaction of the oxide of Na with water (forms NaOH — basic) and the equation for SO2 dissolving in water (forms H2SO3 — acidic), annotating which side of period 3 gives basic and which gives acidic oxides.	'Your arrow says reactivity increases — reactivity with what? With water? With oxygen? Be precise.' For early finishers: 'Add what happens when the oxide of sodium dissolves in water. What does that tell the water officer about what is left behind in the beaker after the explosion?' [WAIT TIME: 5 minutes of quiet individual work.] In the final 2 minutes, cold-call one student to share their annotated table under a document camera or by describing it aloud. Close with: 'Next lesson, we take this model out of the lab and into Kenya — what are these elements actually used for, and why does their chemistry make them useful?'	immediate visual gradient that mirrors the left-to-right ionisation energy trend.	systematic errors in equations written in the model.

D. TEACHER REFLECTION

After the lesson, ask yourself: (1) Could students articulate the nuclear charge and shielding argument in their own words, or were they only restating memorised phrases? (2) Did the POE worksheet structure create genuine cognitive tension, or did students treat it as a formality? (3) Were the demonstrations paced so students had adequate silent recording time after each one, or did the observation phase feel rushed? (4) Did any student raise the aluminium passivation question independently — if not, consider adding a prompt card to the Lesson 7 resources. (5) Were equations balanced correctly and peer-checked effectively, or do most students need additional practice before Lesson 7 presentation work? (6) How well does the class DQB now reflect the journey from Lesson 1 puzzlement to Lesson 6 explanation — are the remaining questions genuinely oriented toward application rather than basic understanding?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Sodium reacted violently with cold water producing a colourless gas that burned with a pop, leaving an alkaline solution; magnesium reacted very slowly with cold water but vigorously with dilute HCl; aluminium showed no visible reaction with cold water; magnesium and sulfur burned brightly in oxygen; the vigour of reaction decreased from left to right across period 3.
What did I learn?	The reactivity of period 3 elements with water and acids decreases across the period because nuclear charge increases while shielding remains nearly constant, raising ionisation energy and making it harder to remove outer electrons; period 3 metals form basic oxides and non-metals form acidic oxides; the products of reactions with water and HCl can be predicted from electron configuration.
How does this explain the phenomenon?	Sodium reacts explosively with Lake Victoria water because it has only one outer electron held by 11 protons with 2 full inner shells providing moderate shielding — its ionisation energy is only 496 kJ per mole and the electron is given up almost instantaneously to water molecules; magnesium has 12 protons with the same inner shielding, so its two outer electrons experience stronger effective nuclear charge and are harder to remove — reactivity is far lower; aluminium and silicon have even higher effective nuclear charges and higher ionisation energies, so they do not react with cold water at all; the anchoring phenomenon is now fully

	explained by the left-to-right increase in effective nuclear charge across period 3.

LESSON 7 (80 minutes): What Are These Elements Good For? — Uses, Applications & Final Explanation	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To consolidate the entire periodicity storyline by connecting electron-configuration-driven trends to real-world Kenyan applications, completing the Driving Question Board, and producing a final annotated periodic-table model that explains the anchoring phenomenon from first principles.
Knowledge	Students will be able to state and explain the uses of selected elements from Groups I, II, VII and VIII in Kenyan contexts (sodium in street lighting/NaOH production, chlorine in water treatment of Lake Victoria water, argon in welding, calcium in cement and construction, iodine in medical antiseptics and nutrition), linking each use directly to the chemical or physical property that makes the element suitable, and articulating how electron configuration underlies those properties.
Skills	Students will search for information using digital devices and/or print media; prepare and deliver a short group presentation; construct a final cumulative annotated periodic-table model comparing it to their Lesson 1 initial model; and collaboratively answer every Driving Question Board question using evidence drawn from Lessons 1–7.
Attitudes	Students will appreciate that chemistry is not confined to the laboratory — it shapes Kenyan infrastructure, public health, and daily nutrition; they will value every group member's contribution (Unity) and present findings with confidence (Self-efficacy).
Key Inquiry Question	1. Why is the study of chemical families important? 2. How does the position of an element in the periodic table influence its properties?
Purpose in Storyline	This is the capstone lesson of the seven-lesson periodicity arc. Students began by watching sodium explode in Lake Victoria water and asking 'why?' Over six lessons they built the conceptual tools — electron configuration, shielding, ionisation energy, electronegativity, atomic/ionic size, and periodic trends — to answer that question rigorously. In Lesson 7 they apply that framework to real Kenyan uses of elements, then return to the Driving Question Board to systematically close every open question, and finally revise their annotated periodic-table model into a complete predictive and explanatory framework, demonstrating full storyline closure.
Safety Notes	No hazardous chemicals are used in this lesson. If digital devices are used, ensure safe and supervised internet access in line with school policy. Print resources should be pre-vetted by the teacher. Students handling printed periodic table models should be reminded to handle scissors or craft materials safely.

B. LESSON OVERVIEW

Lesson 7 is the Final Explanation lesson that closes the seven-lesson periodicity storyline. Students begin by briefly revisiting their Lesson 1 initial models of the sodium-water explosion phenomenon to activate prior learning. They then work in expert groups to research Kenyan applications of elements from Groups I, II, VII and VIII using digital devices or print resources, preparing short presentations that explicitly link each use to an underlying periodic property (e.g., chlorine's high electronegativity and oxidising power make it effective for disinfecting Lake Victoria water at the Kisumu Water and Sanitation plant). Groups present to the class, and the teacher facilitates a whole-class synthesis that weaves every application back to electron configuration. The class then conducts a structured DQB Closure session, systematically reading and answering each question posted in Lesson 1, citing specific evidence from Lessons 1–7. Finally, each student individually revises their annotated periodic-table model, annotating electron configurations, trends, and applications in an integrated diagram, and writes a written Final Explanation answering the driving question. The lesson ends with a class reading of the driving question and a collective affirmation of their answer — a moment of intellectual closure connecting back to the Kisumu demonstration that started the unit.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Periodicity of algebraic models Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Periodicity of Chemical families Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Students individually re-read the anchoring phenomenon description (the Kisumu water purification officer's sodium-water demonstration) displayed on the board. They then silently compare their current understanding to their Lesson 1 initial model, writing two sentences in their notebooks: (1) 'What I thought was happening in Lesson 1:' and (2) 'What I now know is actually happening:'. Students share with a shoulder partner for 2 minutes. The teacher cold-calls 3 pairs to share contrasts. Students then receive a printed copy of all DQB questions generated in Lesson 1 and spend 2 minutes individually marking each question: a tick if they feel confident they can now answer it, a question mark if they are still unsure.	Display the anchoring phenomenon text and the Lesson 1 initial model template on the board simultaneously. Say: 'We started this unit watching sodium explode in Lake Victoria water and you had no idea why. Look at your very first model from Lesson 1. What did you think was happening then?' [WAIT 10 seconds]. Cold-call 3 students. Then say: 'In the next 70 minutes, we are going to prove — using everything from Lessons 1 to 6 — that we can now fully answer the driving question. Let's begin by checking our DQB.' Distribute printed DQB question sheets. After students mark their confidence, tally the class: 'How many questions does everyone feel they can answer? How many still have question marks?' Use this as a live formative checkpoint to decide which DQB questions need the most attention during Phase 4.	Strategy: Prior Knowledge Activation via Model Comparison. Students juxtapose their naïve Lesson 1 model with their current understanding, making cognitive growth explicit and metacognitive. Tallying DQB confidence creates a class-level map of residual uncertainty that guides Phase 4 instruction.	Observable indicator: Every student has written two sentences contrasting Lesson 1 thinking with current understanding. Teacher circulates and checks for specificity — students should reference electron configuration, shielding, or ionisation energy in their 'now I know' sentence, not just vague statements like 'I know more.' Students who write only vague statements are flagged for targeted support during Phase 3.
Observe Phase 📹 **VIDEO:** Periodicity of algebraic models Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Periodicity of Chemical families Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar	The class is divided into four expert groups, each assigned one element group: Group A — Group I elements (sodium: street lighting in Nairobi, NaOH in soap-making, lithium in batteries used in mobile phones across Kenya); Group B — Group II elements (calcium: cement production for Kenyan construction, magnesium in chlorophyll supporting maize and tea farming in the Rift Valley, calcium in milk and bone health); Group C — Group VII elements (chlorine: water treatment at Kisumu Water and Sanitation Company, iodine in antiseptics	Assign groups and post the three-heading framework on the board. Say: 'You have exactly 10 minutes to become the class experts on your group's elements. Your presentation must answer three questions — what is it used for in Kenya, what property makes it useful, and how does electron configuration explain that property. Every single application must connect back to our periodic table model.' [WAIT 12 seconds for groups to orient]. Circulate actively during research — spend approximately 2 minutes with each group. Prompt Group A: 'Why does	Strategy: Jigsaw Expert Teaching. Each group develops deep expertise on one application domain and then teaches the rest of the class, distributing cognitive load while ensuring all students encounter all four groups' applications. Requiring the three-heading structure forces students to move beyond factual recall to mechanistic explanation rooted in electron configuration, directly advancing the driving question.	Observable indicator: Each group's presentation explicitly names a property (e.g., 'chlorine's high electronegativity / oxidising power'), states a Kenyan application, and links the property to electron configuration (e.g., 'chlorine has 7 valence electrons, needs one more to complete its shell, so it strongly pulls electrons from bacterial cell membranes, destroying them'). Groups that present only facts without property-to-configuration links are immediately prompted by the teacher: 'That's a great use — now why does that element have that

readings	used in Kenyan hospitals and in iodised salt added to ugali); Group D — Group VIII elements (argon: inert shielding gas in welding used in infrastructure projects like the SGR, neon in advertising signs in Nairobi's CBD, helium in weather balloons used by the Kenya Meteorological Department). Each group uses digital devices or print resources (teacher-prepared information sheets as fallback) to research their assigned elements for 10 minutes, then prepares a 3-minute oral presentation on a sheet of paper structured around three headings: (1) What is this element used for in Kenya? (2) Which property makes it suitable? (3) How does electron configuration explain that property? Groups present sequentially (12 minutes total, 3 minutes per group).	sodium emit orange-yellow light? Think about energy levels.' Prompt Group B: 'Why is calcium important in cement — what chemical property links to its two valence electrons?' Prompt Group C: 'Why does chlorine kill bacteria in water — link to its electron affinity and oxidising power.' Prompt Group D: 'Why is argon the perfect welding shield — link to its full outer shell and zero reactivity.' During presentations, stand to the side and visually track which DQB questions are being answered. After each presentation, ask one targeted question to the presenting group and one to the listening class. Cold-call at least 2 students per presentation with: 'Can someone from a different group add the electron-configuration explanation that links to what Group [X] just said?'		property? Go back to its electron configuration.'
Explain Phase 📹 **VIDEO:** Periodicity of algebraic models Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Periodicity of Chemical families Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	The teacher facilitates a whole-class synthesis discussion using a large printed or projected periodic table on which key elements (Li, Na, K, Mg, Ca, Cl, Br, I, Ar, Ne) are highlighted. Students contribute to building a class master explanation by responding to a sequence of escalating synthesis questions. The teacher writes student responses on the board in an organised framework under two headings: 'Property' and 'Electron Configuration Root Cause'. Students copy the completed framework into their notebooks. The synthesis culminates in students crafting a class consensus answer to the driving question — written on the board word-by-word as students dictate it and the teacher scribes.	Open with: 'Every application your groups just described — from chlorine in Kisumu's water treatment plant to argon protecting welders on SGR construction sites — has one master explanation. What is it?' [WAIT 15 seconds]. Cold-call 3 students. Guide toward: 'electron configuration determines chemical behaviour, and position in the periodic table tells us the electron configuration.' Then ask the sequence: (1) 'Why does sodium explode in Lake Victoria water but argon does nothing?' — [WAIT 10 sec] — cold-call 2 students; (2) 'Why does potassium explode more violently than sodium, which explodes more violently than lithium?' — [WAIT 12 sec] — cold-call 2 students; (3) 'Why does chlorine disinfect water but argon cannot?' — [WAIT 10	Strategy: Structured Whole-Class Socratic Synthesis. The teacher uses escalating synthesis questions to guide students from isolated facts (uses of elements) to the unifying mechanism (electron configuration as master predictor). Scribing the consensus answer publicly externalises collective understanding, models scientific explanation-writing, and creates an anchor text students can use in their individual Final Explanation.	Observable indicator: The class consensus answer on the board includes all five required elements — electron configuration, period/group position, shielding, ionisation energy, and a Kenyan example. The teacher assesses individual understanding by cold-calling students who were quiet during Phase 2 to contribute at least one clause to the consensus answer. Students who cannot contribute are paired with a peer for support during Phase 5.

		sec] — cold-call 2 students; (4) 'So — using electron configuration — can we predict how any element in the periodic table will behave before we put it in a test tube?' — [WAIT 15 sec] — cold-call 3 students. Scribe student answers into the two-column framework on the board. Then say: 'We are going to write our class answer to the driving question together. I will scribe exactly what you say.' Facilitate word-by-word construction of the consensus answer, ensuring it references: electron configuration, period/group position, shielding, ionisation energy, and at least one Kenyan example.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Periodicity of algebraic models](#) Source: Khan Academy (English - US curriculum) 🔍 [Search ARES for similar videos](#) 📖 **PDF:** [Periodicity of Chemical families](#) Source: Kenya Curriculum Tools 2025 🔍 [Search ARES for similar readings](#)	The teacher projects or displays the full DQB from Lesson 1 (all student questions posted on sticky notes or a digital board). Working in pairs, students receive a numbered list of all DQB questions. Each pair is assigned 2–3 questions to answer in writing using evidence from specific lessons (e.g., 'We learned this in Lesson 3 when we measured ionisation energies across Period 3'). After 5 minutes of pair writing, the teacher conducts a rapid whole-class DQB review: one pair reads their question aloud, states their evidence-based answer, and nominates another pair to confirm, challenge or extend. Every DQB question is addressed. Questions the class cannot yet fully answer are explicitly acknowledged ('This question — [X] — we have partially answered; here is what we know and here is what further chemistry study will reveal'). The teacher physically moves each sticky note from 'Open Questions'	Display the full DQB. Say: 'In Lesson 1, you posted these questions because the sodium explosion puzzled you. Seven lessons later — let's close every single one.' Assign pairs and distribute the numbered DQB question sheets. Circulate during the 5-minute pair-writing phase, checking that answers cite lesson evidence (not just general knowledge). After 5 minutes, begin the rapid review: 'Pair 1 — read Question 1 and give your evidence-based answer.' After each pair answers, ask the class: 'Does anyone want to add evidence from a different lesson?' [WAIT 8 seconds]. Cold-call one additional student per question. As each question is answered, dramatically move the sticky note to the 'Answered' column and say: 'That question is closed!' If any question cannot be fully answered, say: 'We have a partial answer — this is the frontier of what Grade 10 periodicity can explain. In future	Strategy: Evidence-Cited DQB Closure. Requiring pairs to cite specific lesson evidence (not just state answers) reinforces that scientific understanding is built on cumulative evidence. The physical act of moving questions from 'Open' to 'Answered' makes the narrative arc of the unit visible and gives students a concrete sense of intellectual accomplishment, reinforcing the metacognitive dimension of science learning.	Observable indicator: Every pair produces written answers that cite at least one specific lesson by number as evidence (e.g., 'In Lesson 4 we found that ionisation energy increases across Period 3, which explains why...'). The teacher collects these written answer sheets at the end of the lesson as a written formative record. Questions that multiple pairs struggle to answer are flagged for brief teacher clarification before the end of the DQB session.

	to 'Answered Questions' on the DQB, making the intellectual closure visible and celebratory.	chemistry, you will go deeper.' End the DQB session by reading the full driving question aloud with the class and asking: 'Can we now answer this — fully, using evidence?' [WAIT 10 seconds]. Lead the class in a collective 'Yes' and transition to Phase 5.		
Model Building Phase 📹 **VIDEO:** [Periodicity of algebraic models] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** [Periodicity of Chemical families] Source: Kenya Curriculum Tools 2025 🔍 Search ARES for similar readings	Each student individually constructs their Final Cumulative Annotated Periodic Table Model. They begin with a printed partial periodic table (Groups I, II, VII, VIII and Period 3 highlighted) and annotate it with: (1) electron configurations for Li, Na, K, Mg, Ca, Cl, Br, I, Ar; (2) directional trend arrows for atomic size, ionisation energy, electronegativity, and reactivity across Period 3 and down Groups I and VII, with brief written justifications citing shielding and nuclear charge; (3) at least one Kenyan real-world application per group labelled on the relevant element; (4) a 'Phenomenon Explanation Box' in the corner that answers in 3–4 sentences: 'Why does sodium explode violently in Lake Victoria water, while magnesium barely fizzes and argon does nothing?' Students then write a 4–6 sentence Individual Final Explanation answering the driving question in full, referencing evidence from at least three different lessons. Students who finish early annotate additional Period 3 elements (Al, Si, P, S) with their trends. The lesson concludes with 3 volunteers reading their Final Explanation aloud, the teacher highlighting elements of strong scientific explanation in each, and a whole-class acknowledgement of the	Distribute printed partial periodic tables and say: 'This is your final model — the model that explains everything we have seen from Lesson 1 to today. It must have all four layers of annotation I am posting on the board.' Post the four annotation requirements visibly. Say: 'You have 15 minutes. Work in silence — this is your individual scientific record.' Circulate continuously, checking for: correct electron configurations (prompt: 'Sodium is in Period 3, Group I — how many electrons in each shell?'), correctly oriented trend arrows (prompt: 'Is ionisation energy going up or down as you go left to right across Period 3? Why — think nuclear charge'), and Kenyan application labels (prompt: 'Which Kenyan use did your expert group present? Add it here'). After 15 minutes, say: 'Now write your Individual Final Explanation — 4 to 6 sentences answering the driving question. Use the word electron configuration, use the word shielding, use at least one Kenyan example, and explain the sodium explosion.' [WAIT 5 minutes for writing]. Cold-call 3 volunteers to read aloud. For each, highlight one strength: 'Notice how [student name] linked ionisation energy to shielding — that is the heart of the explanation.' End with: 'Seven lessons ago, you watched sodium explode in Lake Victoria water and	Strategy: Cumulative Model Revision as Final Explanation. The annotated periodic table model is a physical, visible record of the student's entire conceptual journey. Requiring four layers of annotation (electron configurations, trend arrows with justifications, Kenyan applications, and the phenomenon explanation box) ensures the model integrates declarative knowledge, causal mechanisms, and real-world relevance into one coherent artefact. The Individual Final Explanation forces students to synthesise this into prose, practising the science practice of Constructing Explanations from Evidence (NGSS SEP 6).	Observable indicators: (1) The annotated model correctly shows electron configurations for all 9 named elements; (2) all four trend arrows are correctly oriented with written justifications referencing at least one of: nuclear charge, shielding, number of electron shells; (3) at least one Kenyan application is labelled per group; (4) the Phenomenon Explanation Box correctly identifies low ionisation energy (due to shielding and one valence electron) as the reason sodium reacts violently, and full outer shell as the reason argon does not react; (5) the Individual Final Explanation includes the words 'electron configuration', 'shielding', and a Kenyan example. Teacher collects all Final Explanation paragraphs as a summative-adjacent formative record of unit-level understanding.

	completed learning journey from puzzled observers of a sodium explosion to confident predictors of elemental behaviour.	had no idea why. Today you can explain it, predict it, and connect it to water treatment, construction, and street lighting across Kenya. That is what chemistry does.' [Pause 10 seconds in silence for effect].		

D. TEACHER REFLECTION

Reflect on the following questions after delivering this lesson:

1. STORYLINE CLOSURE: Did every student's Final Explanation explicitly connect the sodium-water explosion back to electron configuration, shielding, and ionisation energy — or did some students still describe the phenomenon without mechanistic explanation? If explanations remained descriptive, consider building in a structured sentence-starter scaffold in future: 'Sodium reacts violently because its electron configuration ([2,8,1]) means its single outer electron is shielded by 10 inner electrons, giving it a low ionisation energy of only 496 kJ/mol, which means...'

2. DQB CLOSURE QUALITY: Were there DQB questions from Lesson 1 that the class could not answer by Lesson 7? If so, which conceptual gaps do these represent? Are there lessons (Lessons 2–6) where the relevant content was under-emphasised? Note these for curriculum pacing revision.

3. EXPERT GROUP PRESENTATIONS: Did all four expert groups successfully make the property-to-electron-configuration link in their presentations, or did some groups present only factual uses without the mechanistic link? Groups that presented only facts may indicate that the three-heading framework needs to be modelled explicitly with an example before group work begins in future iterations.

4. KENYAN CONTEXT INTEGRATION: Did students spontaneously use Kenyan contexts (Kisumu water treatment, SGR welding, Nairobi street lighting, Rift Valley agriculture) in their Final Explanations, or only when prompted? Strong integration indicates that the Kenya-contextualised approach across all seven lessons has built genuine contextual relevance. Weak integration suggests more explicit prompting is needed in Phases 2–3.

5. CORE COMPETENCY — SELF-EFFICACY: During presentations, did all students present with confidence, or were some hesitant? Note which students struggled to present and consider how to build presentation confidence earlier in the unit's next iteration (e.g., structured peer-practice rounds before the final presentation).

6. PCI — SOCIO-ECONOMIC LINK: Did the discussion of street lighting and road safety (sodium vapour lamps in Nairobi) make the chemistry feel genuinely relevant to students' lives? If students expressed curiosity about other Kenyan infrastructure applications, note these as enrichment threads for future lessons.

7. UNIT-LEVEL REFLECTION: Across all seven lessons, did the anchoring phenomenon (the sodium explosion at the Kisumu Water and Sanitation Company) serve as a sufficiently compelling and recurrent thread? Did students reference it voluntarily throughout the unit, or only when prompted? This is a key indicator of whether the phenomenon-based storyline approach built genuine sustained curiosity across the sub-strand.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students observed (across the unit): sodium metal exploding violently in Lake Victoria water, skimming and igniting with an orange

	flame; magnesium barely fizzing in cold water; aluminium showing no reaction with cold water; and argon being completely inert. They observed a clear left-to-right decrease in reactivity across Period 3 and a clear top-to-bottom increase in reactivity down Group I (lithium < sodium < potassium). In this lesson, students observed expert group presentations revealing that these same reactivity differences directly determine real-world uses — chlorine's reactivity makes it a water disinfectant, argon's inertness makes it a welding shield, sodium's low ionisation energy underlies both its explosive water reaction and its use in sodium vapour street lamps.
What did I learn?	Students learned that an element's position in the periodic table — specifically its group and period number — determines its electron configuration, which in turn determines every physical and chemical property observed. Going down Group I (Li → Na → K), increasing electron shells increase shielding, reducing the effective nuclear charge felt by the outer electron, lowering ionisation energy and increasing reactivity with water. Going across Period 3 (Na → Mg → Al), increasing nuclear charge with the same shielding effect increases ionisation energy and decreases reactivity with cold water. Group VIII elements have full outer shells (noble gas configuration), giving them zero tendency to gain or lose electrons, making them completely inert. These principles explain every Kenyan application studied: chlorine's 7 valence electrons drive its oxidising disinfectant power; argon's full shell makes it the ideal inert welding atmosphere; calcium's 2 valence electrons and moderate reactivity make it suitable for construction chemistry in cement.
How does this explain the phenomenon?	Students can now explain the anchoring phenomenon in full: The water purification officer's sodium explodes in Lake Victoria water because sodium's electron configuration [2,8,1] — with a single outer electron shielded by 10 inner electrons — gives sodium an exceptionally low first ionisation energy (496 kJ/mol). This outer electron is so easily removed that sodium reacts spontaneously and violently with water molecules, releasing hydrogen gas (which ignites) and forming strongly alkaline NaOH. Magnesium [2,8,2] has two outer electrons held by a nuclear charge of +12 but shielded by only 10 inner electrons — a much higher effective nuclear charge — so its ionisation energy is higher and it reacts only slowly with cold water. Argon [2,8,8] has a completely full outer shell with no tendency whatsoever to donate, accept, or share electrons, making it perfectly inert — exactly why the officer uses it to shield sodium during transport. Potassium [2,8,8,1] has its outer electron in a fourth shell, further from the nucleus and shielded by 18 inner electrons, so its ionisation energy is even lower than sodium's, making it explode more violently. Position in the periodic table therefore allows chemists — and Kenyan water treatment engineers, construction workers, and medical professionals — to predict exactly how an element will behave before it ever enters a test tube.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.